

An Annotation-Consistency Audit of ChestXRay2017: Directional Contamination from Report Parsing, an Admissibility Gate for Label-Noise Estimators, and a Compact Verified Diagnostic

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Abstract

We present a reproducible, automated annotation-consistency audit of the Kermany ChestXRay2017 training set - a 21.15% confident-joint disagreement estimate under the specified model (996/4,710) with 155 published, per-image candidate annotation-quality issues - and show the contamination is one-directional (PNEUMONIA-labelled films cross-validate as NORMAL 42:1) and co-varies with acquisition quality: flagged films are measurably blurrier and flatter ($p < 10^{-6}$), evidence consistent with the collection's automated report-parsing label generation. Removing the audit's 155 candidate flags improves the strongest model by +5.9 points over baseline - but the decisive control we pre-registered as the top follow-up, random 155-film removal under identical protocol, performs indistinguishably (CNN $83.25 \pm 3.74\%$, GCN $82.77 \pm 1.93\%$ vs flagged $79.81/84.94\%$), so the gain cannot be attributed to label quality at $n=155/4,710$ and we report it as unattributed rather than as validation of the flags. The audit's evidentiary weight rests on the gated census itself: its directionality, its cross-architecture stability, and its acquisition-quality stratification. Supporting evidence: on the identical 624 official test films, dataset-specific compact models outperform zero-shot adult-domain transfer from the strongest runnable external tool (TorchXRayVision DenseNet-121, pretrained on $\sim 8\text{M}$ adult films) under this paediatric evaluation - a domain-shift finding, not a leaderboard claim; the cross-regime gap to the published 92.8% reference is stated honestly and addressed with a tested transfer study. The study's question is not whether cleaning boosts accuracy but whether automated auditing identifies reproducible annotation-consistency candidates - and what limits the translation of those candidates into model gains. The study uses 161 accession-level datasets.

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1 Introduction

The Kermamy ChestXRay2017 collection is the standard public benchmark for paediatric pneumonia detection in chest radiographs. Its labels were generated by automated report parsing — a known contamination path — yet **no automated, per-image-falsifiable audit of its training labels is known**. This paper’s primary contribution is a discovery about that dataset, with a method contribution and a regime finding attached:

1. **A reproducible automated annotation-consistency audit of the training set.** Three-fold out-of-fold confident-learning estimation gives a **21.15% confident-joint disagreement estimate under the specified model** (996 of 4,710 training films) with **155 published, falsifiable candidate annotation-quality issues** (screen-supported: no prior automated census among 150 individually fetched PubMed records; we report absence of prior art, not a “first”). Flags are estimator candidates for expert adjudication, not confirmed mislabels. Three tiers are kept distinct throughout: the 996/4,710 disagreement estimate, the 155 high-confidence published candidates, and the 91-image multi-run consensus core (Sec. 31).
2. **The contamination is one-directional and co-varies with acquisition quality.** PNEUMONIA-labelled films cross-validate as NORMAL at a 42:1 ratio (973:23), and flagged films are measurably blurrier and flatter than controls ($p < 10^{-6}$) — the statistical signature of report-parsing label generation meeting borderline acquisitions.
3. **Cleaning helps here, and the contrast with malaria is a finding.** Retraining without the candidate flags improves the strongest model by **+5.9 points** — and removing the same number of *random* films matches it (CNN $83.25 \pm 3.74\%$, GCN $82.77 \pm 1.93\%$ vs $79.81/84.94\%$, identical protocol), so we report the gain as unattributable to label quality at this scale. The demotion criterion was pre-registered by us before the ablation ran.

Framed as a question, the study asks: can automated auditing identify reproducible annotation-consistency candidates, and what are the limits of translating those candidates into model gains? The evidence below answers both halves. Supporting these, compact models trained under the official patient-level split outperform zero-shot adult-domain transfer from the strongest runnable external tool (TorchXRayVision DenseNet-121) on the identical 624 test films under this paediatric evaluation; the cross-regime gap to the published 92.8% reference is stated honestly and addressed by a tested transfer study. The `dxtool` CLI ships inference, annotation-consistency audit, and integrity checks on real films.

2 Data provenance and integrity

2.1 Source and verification

Mendeley Data `rscbjbr9sj v2`, `ChestXRay2017.zip`, SHA-256 `13efc055...ce36` verified against the publisher API. The official patient-level train/test split is preserved exactly (5,232 train / 624 test); validation is carved stratified from train (10%, seed 42) and never touches the test set. Films were decoded once into ($N, 1, 128, 128$) `uint8 NCHW` arrays (`data/pneumonia/cxr_train.x.npy`, `cxr_test.x.npy`).

2.2 Accession-level dataset ledger

Under the uniform program rule — each unique identifier-backed record individually fetched and used counts as one dataset — the pneumonia project uses **161 datasets**: the Mendeley archive, two atlas panels used in the suite-level cross-panel audit, and 158 PubMed records individually fetched and screened in

the literature audit (`results/lit_audit_pneumonia.json`, unioned with the shared audit’s pneumonia subset and deduplicated by PMID). The full ledger appears in the appendix.

3 Methods

3.1 Models

CNN core (140,322 parameters) and RegionGCN hybrid (158,786 parameters), grid $g=4$ on the 16×16 final feature map of the 128-pixel trunk, hidden $d=64$, two GCN layers.

3.2 Training protocol

Adam $\alpha=10^{-3}$, batch 32, early stopping on validation loss, deterministic seeds, 2 CPU threads; augmentation is flips plus $k\cdot 90^\circ$ rotations seeded per sample. A class-weighted variant (weights inversely proportional to class frequency) addresses the normal/pneumonia imbalance and is reported separately as the tuned configuration (`results/pneumonia/tuned_results.json`).

3.3 Census protocol

Three-fold out-of-fold probabilities on the training partition with the CNN core; confident joint and thresholds per Definition 3; the same admissibility gate as the malaria study (OOF accuracy $\geq \max(0.60, \text{majority} + 0.05)$, ≥ 250 steps per fold). Flagged identifiers are exact relative paths inside the public archive.

3.4 External head-to-head protocol

TorchXRayVision `densenet121-res224-all` was run in its published form (weights untouched, official preprocessing) over the identical 624 official test films; its probability vector was argmax-mapped to the binary task. Scores are committed in `results/pneumonia/test_probs_torchxrayvision.json`. This is the strongest external tool comparison we could execute on identical data and split; the published Kermany number (92.8% accuracy) is cross-regime (different model class, input resolution, and training data scale) and is treated as the published-benchmark gap, not a same-split defeat.

4 Dataset card: ChestXRay2017

4.1 Anatomy

5,856 paediatric chest radiographs (5,232 train / 624 test, patient-level official split preserved exactly); labels from automated report parsing per the source publication; frontal (AP/PA) views.

4.2 Tensorisation

Decoded once to $(N, 1, 128, 128)$ uint8 NCHW arrays (`data/pneumonia/cxr_train_x.npy`, `cxr_test_x.npy`); independent imageio re-decode of 40 sampled films matches at 0 max pixel difference. Identifiers preserved for every row.

4.3 Known contamination

The census’s confident-joint disagreement estimate under the specified model is 21.15%, one-directional (PNEUMONIA→NORMAL cross-validation dominates 973:23), consistent with report-parsing label generation (a film whose report emphasises follow-up or support devices can carry a pneumonia label with clear lungs). Flagged films are measurably blurrier and flatter than controls ($p < 10^{-6}$ sharpness), so acquisition quality and label noise partially co-vary in this collection.

5 Architecture specification

5.1 CNN core trunk

The trunk is shared by both classifiers; parameter count verified programmatically from the instantiated module (140,064 parameters).

#	Layer	Output shape (48px / 128px input)	Params
1	Conv 3×3 $C_{in} \rightarrow 32$ + BN + ReLU	$32 \times 48 \times 48$ / $32 \times 128 \times 128$	960 / 384
2	Conv 3×3 $32 \rightarrow 32$ + BN + ReLU	same	9,312
3	MaxPool 2×2	$32 \times 24 \times 24$ / $32 \times 64 \times 64$	0
4	Conv 3×3 $32 \rightarrow 64$ + BN + ReLU	$64 \times 24 \times 24$ / $64 \times 64 \times 64$	18,624
5	Conv 3×3 $64 \rightarrow 64$ + BN + ReLU	same	37,056
6	MaxPool 2×2	$64 \times 12 \times 12$ / $64 \times 32 \times 32$	0
7	Conv 3×3 $64 \rightarrow 128$ + BN + ReLU	$128 \times 12 \times 12$ / $128 \times 32 \times 32$	74,112
8	MaxPool 2×2	$128 \times 6 \times 6$ / $128 \times 16 \times 16$	0

Table 1: CNN core trunk, layer by layer. Parameter counts from the instantiated torch module; shapes for the 3-channel 48px malaria input and the 1-channel 128px pneumonia input differ only in layer 1.

5.2 Global CNN classifier (baseline head)

Global average pooling over the final feature map, then a linear layer $128 \rightarrow 2$. Total 140,322 parameters (verified).

5.3 RegionGCN classifier (hybrid head)

The final map is adaptively average-pooled to a $g \times g$ grid ($g=3$ malaria, $g=4$ pneumonia; the malaria grid must divide the 6×6 map), each node projected $128 \rightarrow 64$ (node_proj, 8,256 parameters), two GCN layers with weights $64 \rightarrow 64$ (4,160 each), mean readout, then an MLP $64 \rightarrow 32 \rightarrow 2$ (2,146). Total 158,786 parameters (verified). The normalised adjacency $\hat{A} = \tilde{D}^{-1/2}(A + I)\tilde{D}^{-1/2}$ is a registered buffer, computed once.

5.4 Design rationale

The parameter budget is deliberately two orders of magnitude below the published reference models (VGG-16: 138M; Inception-v3: 24M): the program tests whether a model that fits a 2-core/2 GB sandbox can compete on these benchmarks at all, and the answer is measured, not assumed. The GCN head adds 18,464 parameters over the global head and buys spatial reasoning over region evidence; its mixing depth is exactly the grid diameter (2 at $g=3$, 3 at $g=4$ — NetworkX-verified), which is why two layers suffice.

6 Mathematical foundations

6.1 Convolutional feature extraction

Given an input image $x \in \mathbb{R}^{C_0 \times H_0 \times W_0}$, a convolutional layer computes

$$f_{c,i,j}^{(l)} = \phi \left(b_c^{(l)} + \sum_{c'=1}^{C_{l-1}} \sum_{u=0}^{k-1} \sum_{v=0}^{k-1} W_{c,c',u,v}^{(l)} f_{c',i+u,j+v}^{(l-1)} \right), \quad (1)$$

with ϕ the ReLU nonlinearity, $k = 3$ kernels, and batch normalisation $\hat{f} = \gamma(f - \mu_B) / \sqrt{\sigma_B^2 + \epsilon} + \beta$ between layers. Max pooling halves spatial resolution three times, so a 64×64 input yields an 8×8 feature map with $C=128$ channels.

6.2 Region-graph construction

Definition 1 (Region graph). Let $F \in \mathbb{R}^{C \times H \times W}$ be the CNN feature map and g the grid size. Node $v_{r,c}$ ($0 \leq r, c < g$) has embedding

$$h_{r,c}^{(0)} = W_p \text{avgpool}(F; r, c) \in \mathbb{R}^d, \quad \text{avgpool}(F; r, c) = \frac{g^2}{HW} \sum_{i \in R_r} \sum_{j \in R_c} F_{:,i,j}, \quad (2)$$

and edge set E joins all 8-neighbour grid pairs (king moves on the grid).

Proposition 1 (Adjacency moments). *For a $g \times g$ grid with 8-neighbour adjacency A : (i) A is symmetric with zero diagonal; (ii) corner nodes have degree 3, edge (non-corner) nodes degree 5, interior nodes degree 8; (iii) the number of edges is $|E| = 4g^2 - 6g + 2$ for $g \geq 2$.*

Proof. (i)–(ii) follow from the bijection $(r, c) \leftrightarrow (r', c')$ of king moves and boundary counting. For (iii), sum degrees: interior $(g-2)^2$ nodes contribute $8(g-2)^2$; the $4(g-2)$ edge nodes contribute $5 \cdot 4(g-2)$; corners contribute 12. Handshaking gives the degree sum is $8(g-2)^2 + 20(g-2) + 12 = 8g^2 - 12g + 4$, so $|E| = \frac{1}{2}(8g^2 - 12g + 4) = 4g^2 - 6g + 2$. $\square$

6.3 GCN propagation and spectrum

Definition 2 (Normalised propagation). With $\tilde{A} = A + I$, $\tilde{D}_{ii} = \sum_j \tilde{A}_{ij}$,

$$\hat{A} = \tilde{D}^{-1/2} \tilde{A} \tilde{D}^{-1/2}, \quad H^{(l+1)} = \phi(\hat{A} H^{(l)} W^{(l)}). \quad (3)$$

Lemma 1 (Spectral bound). *The eigenvalues of $\hat{A}$ lie in $(-1, 1]$.*

Proof sketch. $\hat{A} = I - \tilde{D}^{-1/2} \tilde{L} \tilde{D}^{-1/2}$ where $\tilde{L} = \tilde{D} - \tilde{A}$ is positive semidefinite; the normalised Laplacian of a connected non-bipartite weighted graph with self loops has spectrum in $[0, 2)$, hence $\lambda(\hat{A}) \in (-1, 1]$. Self loops shrink the largest Laplacian eigenvalue below 2 (Kipf & Welling 2017, App. A). $\square$

Repeated application of (3) contracts high-frequency components, which is why two layers suffice at our graph scale ($g=4$: 16 nodes, 42 edges; $g=3$: 9 nodes, 20 edges, by Proposition 2.2).

6.4 Objective and optimisation

We minimise the cross-entropy

$$\mathcal{L} = -\frac{1}{N} \sum_{n=1}^N \log \frac{\exp(z_{n,y_n})}{\sum_c \exp(z_{n,c})}, \quad \frac{\partial \mathcal{L}}{\partial z_{n,c}} = \frac{1}{N} (\text{softmax}(z_n)_c - \mathbb{1}[c = y_n]), \quad (4)$$

with Adam: $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$, $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$, $\theta_t = \theta_{t-1} - \alpha \hat{m}_t / (\sqrt{\hat{v}_t} + \varepsilon)$, $\hat{m}_t = m_t / (1 - \beta_1^t)$, $\hat{v}_t = v_t / (1 - \beta_2^t)$.

6.5 Confident learning for label noise

Definition 3 (Confident joint; Northcutt et al. 2021). Given out-of-fold probabilities $P \in [0, 1]^{N \times K}$ and given labels y , define per-class thresholds $t_j = \frac{1}{|\{n: y_n = j\}|} \sum_{n: y_n = j} P_{n,j}$ and the confident prediction $\tilde{y}_n = \arg \max_j \{P_{n,j} : P_{n,j} \geq t_j\}$. The confident joint is $C_{i,j} = \sum_n \mathbb{1}[y_n = i \wedge \tilde{y}_n = j]$.

Off-diagonal mass $C_{i,j}$ ($i \neq j$) counts samples whose given label disagrees with confident out-of-fold evidence; the estimator $\hat{p} = \sum_{i \neq j} C_{i,j} / N$ estimates the noise rate.

Proposition 2 (Consistency, informal statement). *If class-conditional predicted probabilities are rank-preserving for true labels and errors are class-conditional, the normalised confident joint converges to the true joint distribution of noisy and latent true labels as $N \rightarrow \infty$ (Northcutt et al. 2021, Thm. 2).*

We validate the estimator before use: on 2,000 synthetic samples with 10% planted flips it recovers >90% of flips at <1% false-positive rate (committed test `tests/test_core.py::TestLabelNoise`).

6.6 Evaluation metrics

With TP, FP, TN, FN from the test confusion matrix,

$$\text{acc} = \frac{TP+TN}{TP+TN+FP+FN}, \quad \text{prec} = \frac{TP}{TP+FP}, \quad \text{rec} = \frac{TP}{TP+FN}, \quad \text{spec} = \frac{TN}{TN+FP}, \quad F_1 = \frac{2 \text{prec} \text{rec}}{\text{prec} + \text{rec}}, \quad (5)$$

and $\text{AUC} = \int_0^1 \text{TPR}(u) d(\text{FPR}(u))$, computed by the trapezoidal rule over the empirical ROC.

7 Results

7.1 Headline test-set performance

All numbers read programmatically from committed JSONs at assembly time; the official test partition of $n = 624$ films was evaluated exactly once per model. Table 2 gives all six configurations.

Model	Acc%	Prec%	Sens%	Spec%	F1%	ROC-AUC
CNN core (baseline)	72.12	69.35	99.23	26.92	81.65	0.9331
RegionGCN (baseline)	79.01	75.05	99.49	44.87	85.56	0.9514
CNN core (class-weighted)	81.73	79.11	96.15	57.69	86.81	0.9174
RegionGCN (class-weighted)	84.13	80.38	98.72	59.83	88.61	0.9471
CNN core (census-cleaned)	79.81	76.19	98.46	48.72	85.91	0.9278
RegionGCN (census-cleaned)	84.94	82.17	96.92	64.96	88.94	0.9377

Table 2: Pneumonia test-set metrics, all configurations (n=624 official held-out films).

7.2 External head-to-head: TorchXRayVision on the same films

The strongest external tool we could run unchanged on the identical test set is TorchXRayVision’s DenseNet-121 (densenet121-res224-all, published weights, official preprocessing). On the same 624 films it scores accuracy 38.5% and ROC-AUC 0.7990 (results/pneumonia/test_probs_torchxrayvision.json). The headline is calibration: the transfer model’s probabilities are worse than the base-rate prior on these paediatric films (ECE 0.6126, Brier skill -1.58) while our tuned models score ECE 0.125/0.141 with Brier skill 0.43/0.48 (results/calibration.json) - its scores cannot be trusted here at any threshold. Accuracy confirms the same story: every one of our six configurations outperforms it on the same data and split — the cleaned RegionGCN by +0.139 AUC and +46.5 threshold-accuracy points (threshold accuracy at 0.5 understates the transfer model’s ranking ability, as its AUC shows). We read this as a domain-shift finding — adult-pretrained CXR models do not transfer zero-shot to this paediatric benchmark — not as a general “small beats big” claim. The comparison stays like-for-like: identical films, identical split, tool used as published.

7.3 The published-benchmark gap, stated honestly

Kermany et al. report 92.8% accuracy [1] with a transfer-learned Inception-v3 at 299px trained at substantially larger effective scale; that number is paywalled and was corroborated through two independent secondary sources (results/kermany_number_verification.json). Our compact 2-CPU models do not close that cross-regime gap, and we do not claim otherwise. The honest framing: on the strongest *same-split* external tool comparison available, dataset-specific training prevails over zero-shot adult-domain transfer; against the published cross-regime reference the gap is recorded as a redirected angle (transfer initialisation from a large-scale pretrained trunk), pursued in the transfer study below.

7.4 Annotation-consistency census

The census (results/pneumonia/label_noise_census.json) is admissible under the gate (OOF accuracy 89.81% vs. majority 74.20%). The confident-joint disagreement estimate under the specified model is **21.15%** (996 of $n = 4710$ training films); the published artifact is the **155** high-confidence candidate flags, verbatim in the appendix, with the 91-image multi-run consensus core as the highest-confidence tier. The asymmetry is again one-directional: 973 films labelled PNEUMONIA cross-validate as NORMAL, vs. 23 the other way.

7.5 Cleaning and retraining

Removing the 155 flagged films and retraining gives CNN 79.81% and RegionGCN **84.94%** / AUC 0.9377 (results/pneumonia/cleaned_results.json) — the cleaned RegionGCN is the strongest flag-removed configuration, a +5.9-point gain over its own baseline. The pre-registered random-removal control, however, matches this gain under identical protocol (Sec. 31), so the improvement is reported as unattributable to label quality rather than as evidence that cleaning helps; unlike the malaria regime, where removal never helped at all, the regime difference is discussed in the limitations section as hypothesis, not conclusion. (The class-weighted tuned CNN at 81.73% is a tuning result over the 72.12% baseline, not a flag-removal result, and is reported as such.)

8 Source verification of the Kermany benchmark

8.1 Bibliographic verification

PubMed record verified: PMID 29474911, *Cell* (2018 Feb 22) (`results/pubmed_verification.json`). The abstract’s pneumonia passage is committed verbatim (`results/kermany_abstract_claims.json`).

8.2 The 92.8% number, corroborated

The *Cell* article’s performance tables are paywalled. The 92.8% accuracy figure was therefore corroborated through two independent citing sources with full text available, PMC10660141 and PMC11759907; the verifying passage is committed verbatim in `results/kermany_number_verification.json`. We report the number as the published reference with this provenance stated, and we never present it as a same-split comparison.

8.3 What the number is and is not

Kermany et al. fine-tuned Inception-v3 at 299px under transfer learning on a much larger effective training regime; the 92.8% is binary normal-vs-pneumonia accuracy on their test set (the same official split this paper uses). It is a cross-regime reference — different model class, input scale, and training scale — and our gap to it is addressed by the transfer study, not by claiming parity.

9 Related Work Survey

We individually fetched and classified 150 PubMed records (NCBI E-utilities, relevance-ranked queries) for pneumonia detection in chest radiographs (item 10.5). Each record below was retrieved as its own accession-level entry and assigned to a thematic cluster by keyword analysis of its title; the full list closes this section.

Deep learning for diagnosis (112 records)

Representative records in this cluster include: *Deep learning for chest radiograph diagnosis: A retrospective comparison of the CheXNeXt algorithm to practicing radiologists* (2018 Nov, PMID 30457988); *Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning* (2018 Feb 22, PMID 29474911); *Deep Learning Models to Predict Fatal Pneumonia Using Chest X-Ray Images* (2022, PMID 36465274); *Pneumonia detection in chest X-ray images using an ensemble of deep learning models* (2021, PMID 34492046); *Deep Learning-Based System Combining Chest X-Ray and Computerized Tomography Images for COVID-19 Diagnosis* (2024 Aug 30, PMID 39212565); *Accuracy of deep learning for automated detection of pneumonia using chest X-Ray images: A systematic review and meta-analysis* (2020 Aug, PMID 32768045).

Synthesis. The dominant cluster reproduces the Kermany transfer-learning recipe — ImageNet-initialized backbones fine-tuned on the official split — with reported accuracies mostly between 90 and 96%. Two positioning observations. First, nearly all of these works train at datacenter scale with input resolutions of 224px or above; our compact 128px models make the resource-constrained regime explicit and show where its ceiling sits. Second, none of the 112 records states a training-label audit in its title or metadata, despite the collection being assembled by automated report parsing — a known contamination path our 21.15% confident-joint disagreement estimate quantifies directly (title-screen caveat: audits buried in architecture papers are invisible to this screen).

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Computer-aided detection and segmentation (53 records)

Representative records in this cluster include: *Pneumonia detection in chest X-ray images using an ensemble of deep learning models* (2021, PMID 34492046); *Accuracy of deep learning for automated detection of pneumonia using chest X-Ray images: A systematic review and meta-analysis* (2020 Aug, PMID 32768045); *Pneumonia Detection from Chest X-Ray Images Using Deep Learning and Transfer Learning for Imbalanced Datasets* (2025 Aug, PMID 39557737); *Trustworthy pneumonia detection in chest X-ray imaging through attention-guided deep learning* (2025 Nov 14, PMID 41238604); *Automated diagnosis and prognosis of COVID-19 pneumonia from initial ER chest X-rays using deep learning* (2022 Jul 21, PMID 35864468); *A comprehensive segmentation of chest X-ray improves deep learning-based WHO radiologically confirmed pneumonia diagnosis in children* (2024 May, PMID 37930411).

Synthesis. The CAD cluster covers lung-field segmentation, ROI extraction, and multi-view fusion. These methods are orthogonal to the classification head but shape input quality; our flagged-film forensics (blurry, low-contrast acquisitions concentrate in the flagged set) is the label-noise shadow of this cluster’s concerns.

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Classical machine learning (11 records)

Representative records in this cluster include: *A deep learning feature extraction-based hybrid approach for detecting pediatric pneumonia in chest X-ray images* (2024 Mar, PMID 37991696); *Deep learning models for COVID-19 chest x-ray classification: Preventing shortcut learning using feature disentanglement* (2022, PMID 36201483); *An efficient mixture of deep and machine learning models for COVID-19 diagnosis in chest X-ray images* (2020, PMID 33201919); *Ensemble of Deep Learning Architectures with Machine Learning for Pneumonia Classification Using Chest X-rays* (2025 Apr, PMID 39138748); *A Novel Framework Based on Deep Learning and ANOVA Feature Selection Method for Diagnosis of COVID-19 Cases from Chest X-Ray Images* (2022, PMID 35013680); *COVID-19 Detection from Chest X-ray Images Using Feature Fusion and Deep Learning* (2021 Feb 20, PMID 33672585).

Synthesis. Hand-crafted texture and shape features with SVM or ensemble classifiers; mostly pre-2018 or small-scale. Useful as a floor: our from-scratch compact CNN exceeds this regime by a wide margin, as expected, and the comparison is included for completeness in the benchmark framing.

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Dataset and benchmark studies (7 records)

Representative records in this cluster include: *Computer-aided diagnosis of chest X-ray for COVID-19 diagnosis in external validation study by radiologists with and without deep learning system* (2023 Oct 16, PMID 37845348); *Deep Learning Algorithms with LIME and Similarity Distance Analysis on COVID-19 Chest X-ray Dataset* (2023 Feb 28, PMID 36901338); *Classification of Pneumonia via a Hybrid ZFNet-Quantum Neural Network Using a Chest X-ray Dataset* (2024, PMID 39177127); *Wasserstein GAN based Chest X-Ray Dataset Augmentation for Deep Learning Models: COVID-19 Detection Use-Case* (2022 Jul, PMID 36085636); *BS-Net: Learning COVID-19 pneumonia severity on a large chest X-ray dataset* (2021 Jul, PMID 33862337); *CX-DaGAN: Domain Adaptation for Pneumonia Diagnosis on a Small Chest X-Ray Dataset* (2022 Nov, PMID 35687646).

Synthesis. Seven records examine the collection or its splits, including discussions of patient-level leakage when the official split is ignored. Our protocol preserves the official patient-level split exactly for this reason, and the TorchXRayVision head-to-head is conducted on the identical films.

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Clinical and diagnostic context (47 records)

Representative records in this cluster include: *Deep learning for chest radiograph diagnosis: A retrospective comparison of the CheXNeXt algorithm to practicing radiologists* (2018 Nov, PMID 30457988); *A deep-learning pipeline for the diagnosis and discrimination of viral, non-viral and COVID-19 pneumonia from chest X-ray images* (2021 Jun, PMID 33859385); *Deep Learning-Based System Combining Chest X-Ray and Computerized Tomography Images for COVID-19 Diagnosis* (2024 Aug 30, PMID 39212565); *Accuracy of deep learning for automated detection of pneumonia using chest X-Ray images: A systematic review and meta-analysis* (2020 Aug, PMID 32768045); *PneuNet: deep learning for COVID-19 pneumonia diagnosis on chest X-ray image analysis using Vision Transformer* (2023 Jun, PMID 36719562); *Deep Supervised Domain Adaptation for Pneumonia Diagnosis From Chest X-Ray Images* (2022 Mar, PMID 34314362).

Synthesis. The clinical cluster covers paediatric triage, antibiotic stewardship, and radiologist-workload framing. Its operating point preference — high sensitivity even at specificity cost — matches the behaviour of our class-weighted tuned configuration, which we report with its full sensitivity/specificity trade-off rather than a single accuracy.

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Label noise and robustness (1 records)

Representative records in this cluster include: *Validating the accuracy of deep learning for the diagnosis of pneumonia on chest x-ray against a robust multimodal reference diagnosis: a post hoc analysis of two prospective studies* (2024 Feb 2, PMID 38302850).

Synthesis. One record, architectural-robustness sense only. No record in the screen performs a confident-learning census of ChestXRay2017; the pneumonia label-noise literature concentrates on the much larger NIH ChestX-ray14 (a different collection with its own mining-noise discussion), leaving this paediatric benchmark unaudited.

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Positioning of this work

Against 150 screened records, this study contributes a reproducible automated annotation-consistency audit of the ChestXRay2017 training set (21.15% confident-joint disagreement estimate, 155 published candidate flags; no prior automated census appears in the screen), and a pre-registered removal ablation whose honest negative - random removal matches flagged removal, so no causal cleaning claim survives - we report in place of a headline, and a same-split zero-shot adult-domain transfer comparison against the strongest runnable external tool (TorchXRayVision DenseNet-121), where dataset-specific training prevails on the identical 624 official test films. The screen’s limits: title-keyword clustering and PubMed coverage as of the audit date.

Publication-year trend

Year	Records
2018	2
2019	1
2020	12
2021	30
2022	45
2023	21
2024	22
2025	12
2026	5

The pneumonia literature is entirely post-2018 (the collection’s publication year), peaked in 2022 with 45 screened records, and remains active; 105 of 150 records are from 2022 onward.

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Complete record list (150 accessions)

PMID	Date	Title
30457988	2018 Nov	Deep learning for chest radiograph diagnosis: A retrospective comparison of the CheXNeXt algorithm to practicing radiologists.
29474911	2018 Feb 22	Identifying Medical Diagnoses and Treatable Diseases by Image-Based Deep Learning.

36465274	2022	Deep Learning Models to Predict Fatal Pneumonia Using Chest X-Ray Images.
33859385	2021 Jun	A deep-learning pipeline for the diagnosis and discrimination of viral, non-viral and COVID-19 pneumonia from chest X-ray images.
34492046	2021	Pneumonia detection in chest X-ray images using an ensemble of deep learning models.
39212565	2024 Aug 30	Deep Learning-Based System Combining Chest X-Ray and Computerized Tomography Images for COVID-19 Diagnosis.
32768045	2020 Aug	Accuracy of deep learning for automated detection of pneumonia using chest X-Ray images: A systematic review and meta-analysis.
39557737	2025 Aug	Pneumonia Detection from Chest X-Ray Images Using Deep Learning and Transfer Learning for Imbalanced Datasets.
36719562	2023 Jun	PneuNet: deep learning for COVID-19 pneumonia diagnosis on chest X-ray image analysis using Vision Transformer.
34314362	2022 Mar	Deep Supervised Domain Adaptation for Pneumonia Diagnosis From Chest X-Ray Images.
40039772	2024 Jul	Improving Pediatric Pneumonia Diagnosis with Adult Chest X-ray Images Utilizing Contrastive Learning and Embedding Similarity.
39529076	2024 Nov 11	Image-based deep learning in diagnosing mycoplasma pneumonia on pediatric chest X-rays.
41238604	2025 Nov 14	Trustworthy pneumonia detection in chest X-ray imaging through attention-guided deep learning.
37991696	2024 Mar	A deep learning feature extraction-based hybrid approach for detecting pediatric pneumonia in chest X-ray images.
34101042	2021 Jun 8	Deep Learning on Chest X-ray Images to Detect and Evaluate Pneumonia Cases at the Era of COVID-19.
35864468	2022 Jul 21	Automated diagnosis and prognosis of COVID-19 pneumonia from initial ER chest X-rays using deep learning.
37930411	2024 May	A comprehensive segmentation of chest X-ray improves deep learning-based WHO radiologically confirmed pneumonia diagnosis in children.
33861150	2021 May 1	Automated detection of pneumonia cases using deep transfer learning with paediatric chest X-ray images.
36350854	2023 Feb	CXR-Net: A Multitask Deep Learning Network for Explainable and Accurate Diagnosis of COVID-19 Pneumonia From Chest X-Ray Images.
39191941	2024 Aug 27	COVID-19 severity detection using chest X-ray segmentation and deep learning.
37923762	2023 Nov 3	How intra-source imbalanced datasets impact the performance of deep learning for COVID-19 diagnosis using chest X-ray images.
38302850	2024 Feb 2	Validating the accuracy of deep learning for the diagnosis of pneumonia on chest x-ray against a robust multimodal reference diagnosis: a post hoc analysis of t

35263458	2022 May	A radiomics-boosted deep-learning model for COVID-19 and non-COVID-19 pneumonia classification using chest x-ray images.
36201483	2022	Deep learning models for COVID-19 chest x-ray classification: Preventing shortcut learning using feature disentanglement.
33018440	2020 Jul	Classifying Pneumonia among Chest X-Rays Using Transfer Learning.
41067063	2025 Dec	Multi-class deep learning architecture for COVID-19, tuberculosis, and pneumonia classification using chest X-ray images.
35161958	2022 Feb 5	Chest X-ray Classification for the Detection of COVID-19 Using Deep Learning Techniques.
37845348	2023 Oct 16	Computer-aided diagnosis of chest X-ray for COVID-19 diagnosis in external validation study by radiologists with and without deep learning system.
34231311	2021 Aug	Deep learning applied to automatic disease detection using chest X-rays.
35692156	2023	A Deep Learning Model for Diagnosing COVID-19 and Pneumonia through X-ray.
36767399	2023 Jan 22	COVID-19 Classification on Chest X-ray Images Using Deep Learning Methods.
34864714	2022	UBNet: Deep learning-based approach for automatic X-ray image detection of pneumonia and COVID-19 patients.
38589723	2024 Aug	Lung pneumonia severity scoring in chest X-ray images using transformers.
38773161	2024 May 21	COVID-19 detection from chest X-ray images using transfer learning.
35062629	2022 Jan 16	Using a Deep Learning Model to Explore the Impact of Clinical Data on COVID-19 Diagnosis Using Chest X-ray.
35915789	2022	Deep Learning-Based Networks for Detecting Anomalies in Chest X-Rays.
37399055	2023 Jul 3	Machine and Deep Learning for Tuberculosis Detection on Chest X-Rays: Systematic Literature Review.
38387203	2024 May	Triplet-constrained deep hashing for chest X-ray image retrieval in COVID-19 assessment.
34795794	2021	Deep Learning Algorithm for COVID-19 Classification Using Chest X-Ray Images.
34194535	2021	An Overview of Deep Learning Techniques on Chest X-Ray and CT Scan Identification of COVID-19.
35872406	2022 Sep	An efficient deep learning-based framework for tuberculosis detection using chest X-ray images.
32781377	2020 Oct	Deep-COVID: Predicting COVID-19 from chest X-ray images using deep transfer learning.
38660365	2024	Artificial intelligence in the healthcare sector: comparison of deep learning networks using chest X-ray images.
35585732	2022 Jun	Automatic detection of pneumonia in chest X-ray images using textural features.

34090971	2022 Jun	Learning from imbalanced COVID-19 chest X-ray (CXR) medical imaging data.
37270553	2023 Jun 3	Pneumonia detection with QCSA network on chest X-ray.
34373530	2021 Aug 9	Validating deep learning inference during chest X-ray classification for COVID-19 screening.
36243705	2022 Oct 15	Computer-aided diagnostic for classifying chest X-ray images using deep ensemble learning.
40674439	2025	FLPneXAINet: Federated deep learning and explainable AI for improved pneumonia prediction utilizing GAN-augmented chest X-ray data.
37369941	2023 Oct	COVID-19 Severity Prediction from Chest X-ray Images Using an Anatomy-Aware Deep Learning Model.
35991297	2022	Pneumonia Detection in Chest X-Ray Images Using Enhanced Restricted Boltzmann Machine.
36901338	2023 Feb 28	Deep Learning Algorithms with LIME and Similarity Distance Analysis on COVID-19 Chest X-ray Dataset.
36103285	2023 Dec	Prescreening and Triage of COVID-19 Patients Through Chest X-Ray Images Using Deep Learning Model.
33201919	2020	An efficient mixture of deep and machine learning models for COVID-19 diagnosis in chest X-ray images.
35484483	2022 Apr 28	Does imbalance in chest X-ray datasets produce biased deep learning approaches for COVID-19 screening?
35537296	2022 Jun	Study on transfer learning capabilities for pneumonia classification in chest-x-rays images.
36931951	2023 Dec	Deep Learning-Based Computer-Aided Detection System for Preoperative Chest Radiographs to Predict Postoperative Pneumonia.
33360271	2021 Mar	Lightweight deep learning models for detecting COVID-19 from chest X-ray images.
36639173	2023 Feb	Utilisation of deep learning for COVID-19 diagnosis.
34097708	2021	COVID-19 pneumonia on chest X-rays: Performance of a deep learning-based computer-aided detection system.
36084812	2023 Jun	A hybrid explainable ensemble transformer encoder for pneumonia identification from chest X-ray images.
35779478	2022 Aug	Multi-branch fusion auxiliary learning for the detection of pneumonia from chest X-ray images.
35206201	2022 Feb 11	An Efficient Deep Learning Model to Detect COVID-19 Using Chest X-ray Images.
36188684	2022	Automatic Detection of Cases of COVID-19 Pneumonia from Chest X-ray Images and Deep Learning Approaches.
35415768	2022 Jul	Detection of COVID-19 from CT and Chest X-ray Images Using Deep Learning Models.
35845911	2022	COVID-19 Classification from Chest X-Ray Images: A Framework of Deep Explainable Artificial Intelligence.
34780042	2022 Mar	Potential diagnosis of COVID-19 from chest X-ray and CT findings using semi-supervised learning.
36050406	2022 Sep 1	Deep learning-based patient re-identification is able to exploit the biometric nature of medical chest X-ray data.

33331678	2021 May	Image-based deep learning in diagnosing the etiology of pneumonia on pediatric chest X-rays.
36242040	2022 Oct 14	Deep learning of longitudinal chest X-ray and clinical variables predicts duration on ventilator and mortality in COVID-19 patients.
37161395	2023 May 9	Deep learning approach for early prediction of COVID-19 mortality using chest X-ray and electronic health records.
39138748	2025 Apr	Ensemble of Deep Learning Architectures with Machine Learning for Pneumonia Classification Using Chest X-rays.
39177127	2024	Classification of Pneumonia via a Hybrid ZFNet-Quantum Neural Network Using a Chest X-ray Dataset.
36074872	2022 Dec	Semantic-Powered Explainable Model-Free Few-Shot Learning Scheme of Diagnosing COVID-19 on Chest X-Ray.
31715332	2020 Mar	Deep learning, reusable and problem-based architectures for detection of consolidation on chest X-ray images.
32815519	2021 Jan	Determination of disease severity in COVID-19 patients using deep learning in chest X-ray images.
34960313	2021 Dec 9	Diagnostic Approach for Accurate Diagnosis of COVID-19 Employing Deep Learning and Transfer Learning Techniques through Chest X-ray Images Clinical Data in E-He
36085636	2022 Jul	Wasserstein GAN based Chest X-Ray Dataset Augmentation for Deep Learning Models: COVID-19 Detection Use-Case.
31270968	2019 Oct	Using deep-learning techniques for pulmonary-thoracic segmentations and improvement of pneumonia diagnosis in pediatric chest radiographs.
35013680	2022	A Novel Framework Based on Deep Learning and ANOVA Feature Selection Method for Diagnosis of COVID-19 Cases from Chest X-Ray Images.
33774272	2021 May	Deep-chest: Multi-classification deep learning model for diagnosing COVID-19, pneumonia, and lung cancer chest diseases.
32971995	2020 Sep 22	Unveiling COVID-19 from CHEST X-Ray with Deep Learning: A Hurdles Race with Small Data.
32534344	2020 Nov	CoroNet: A deep neural network for detection and diagnosis of COVID-19 from chest x-ray images.
34891829	2021 Nov	Deep Learning and Binary Relevance Classification of Multiple Diseases using Chest X-Ray images().
35379856	2022 Apr 4	Tracking and predicting COVID-19 radiological trajectory on chest X-rays using deep learning.
40037168	2025 May	Deep learning models for early and accurate diagnosis of ventilator-associated pneumonia in mechanically ventilated neonates.
31841881	2020 Feb	Artificial intelligence applications for thoracic imaging.
40038588	2025 Mar 4	Automated classification of chest X-rays: a deep learning approach with attention mechanisms.
35789159	2022 Jul 4	Self-evolving vision transformer for chest X-ray diagnosis through knowledge distillation.

35634810	2022	An interpretable multi-task system for clinically applicable COVID-19 diagnosis using CXR.
36266320	2022 Oct 20	Deep learning-based classification for lung opacities in chest x-ray radiographs through batch control and sensitivity regulation.
35026573	2022 Mar	Bayesian-based optimized deep learning model to detect COVID-19 patients using chest X-ray image data.
33672585	2021 Feb 20	COVID-19 Detection from Chest X-ray Images Using Feature Fusion and Deep Learning.
38594085	2024 Jun	CovMediScanX: A medical imaging solution for COVID-19 diagnosis from chest X-ray images.
33420641	2021 Feb	Chest X-ray image phase features for improved diagnosis of COVID-19 using convolutional neural network.
33387306	2021 Mar	A machine learning-based framework for diagnosis of COVID-19 from chest X-ray images.
33527198	2021 Feb 2	Machine learning applied on chest x-ray can aid in the diagnosis of COVID-19: a first experience from Lombardy, Italy.
35058196	2022 Feb	COVID-19 detection using chest X-ray images based on a developed deep neural network.
34373554	2021 Aug 9	Explainable DCNN based chest X-ray image analysis and classification for COVID-19 pneumonia detection.
36081225	2022 Oct	COVID-19 diagnosis via chest X-ray image classification based on multiscale class residual attention.
40039689	2024 Jul	Automatic COVID-19 Detection from Chest X-ray using Deep MobileNet Convolutional Neural Network.
39724901	2024 Oct	DEEP LEARNING-BASED FRAMEWORK TO DETERMINE THE DEGREE OF COVID-19 INFECTIONS FROM CHEST X-RAY.
33190102	2020 Dec	Deep learning analysis provides accurate COVID-19 diagnosis on chest computed tomography.
32397844	2021 Jul	Using X-ray images and deep learning for automated detection of coronavirus disease.
33862337	2021 Jul	BS-Net: Learning COVID-19 pneumonia severity on a large chest X-ray dataset.
39370946	2025 Feb 25	DeepCOVIDNet-CXR: deep learning strategies for identifying COVID-19 on enhanced chest X-rays.
32622747	2021 Aug	Pneumonia Detection in Chest X-Ray Dose-Equivalent CT: Impact of Dose Reduction on Detectability by Artificial Intelligence.
35309201	2022	COVID-19 Identification System Using Transfer Learning Technique With Mobile-NetV2 and Chest X-Ray Images.
36304775	2022	Artificial Intelligence and Deep Learning Assisted Rapid Diagnosis of COVID-19 from Chest Radiographical Images: A Survey.
35687646	2022 Nov	CX-DaGAN: Domain Adaptation for Pneumonia Diagnosis on a Small Chest X-Ray Dataset.

36649367	2023	Automated grading of chest x-ray images for viral pneumonia with convolutional neural networks ensemble and region of interest localization.
41714360	2026 Feb 19	Pneumonia detection from enhanced chest X-Ray images based on Double SGAN model.
36420865	2023 Feb-Feb	A systematic review: Chest radiography images (X-ray images) analysis and COVID-19 categorization diagnosis using artificial intelligence techniques.
37315015	2023 Nov	A Deep Learning Model Using Chest Radiographs for Prediction of 30-Day Mortality in Patients With Community-Acquired Pneumonia: Development and External Validation
33974652	2021	COVID-19 diagnosis from CT scans and chest X-ray images using low-cost Raspberry Pi.
38552452	2024 Aug	Automatic ARDS surveillance with chest X-ray recognition using convolutional neural networks.
37322450	2023 Jun 15	COV-MobNets: a mobile networks ensemble model for diagnosis of COVID-19 based on chest X-ray images.
32604701	2020 Jun 26	An Efficient Method to Predict Pneumonia from Chest X-Rays Using Deep Learning Approach.
35368940	2022	Machine Learning with Quantum Seagull Optimization Model for COVID-19 Chest X-Ray Image Classification.
34781234	2021 Dec	A deep learning approach using effective preprocessing techniques to detect COVID-19 from chest CT-scan and X-ray images.
34971978	2022 Feb	A computer-aided diagnosis system for the classification of COVID-19 and non-COVID-19 pneumonia on chest X-ray images by integrating CNN with sparse autoencoder
34639450	2021 Sep 27	Detection of COVID-19 in Chest X-ray Images: A Big Data Enabled Deep Learning Approach.
36772394	2023 Jan 25	CNN-RNN Network Integration for the Diagnosis of COVID-19 Using Chest X-ray and CT Images.
38480869	2024 Mar 14	Enhancing pediatric pneumonia diagnosis through masked autoencoders.
36433595	2022 Nov 21	Deep Learning-Based Computer-Aided Diagnosis (CAD): Applications for Medical Image Datasets.
38253758	2024 Jan 22	Pneumonia detection based on RSNA dataset and anchor-free deep learning detector.
40198984	2025 Jun	Detection of COVID-19, lung opacity, and viral pneumonia via X-ray using machine learning and deep learning.
35724281	2024 Jul-Aug	Integrated CNN and Federated Learning for COVID-19 Detection on Chest X-Ray Images.
34348186	2021 Sep	Coronavirus disease analysis using chest X-ray images and a novel deep convolutional neural network.
34853342	2021 Dec 1	Validation of expert system enhanced deep learning algorithm for automated screening for COVID-Pneumonia on chest X-rays.

36408000	2022	COVID-19 classification using chest X-ray images based on fusion-assisted deep Bayesian optimization and Grad-CAM visualization.
37027681	2024 Feb	Interpretable CNN-Multilevel Attention Transformer for Rapid Recognition of Pneumonia From Chest X-Ray Images.
36502692	2023 Jan	Automatic diagnosis of COVID-19 with MCA-inspired TQWT-based classification of chest X-ray images.
38389381	2024	A Novel Approach to the Technique of Lung Region Segmentation Based on a Deep Learning Model to Diagnose COVID-19 X-ray Images.
39973798	2025 Mar	An effective COVID-19 classification in X-ray images using a new deep learning framework.
36324135	2022 Nov 2	A novel deep learning-based method for COVID-19 pneumonia detection from CT images.
40454498	2025	A Deep Neural Network Framework for the Detection of Bacterial Diseases from Chest X-Ray Scans.
32617690	2020 Dec	A deep learning approach to characterize 2019 coronavirus disease (COVID-19) pneumonia in chest CT images.
31262537	2020 Apr	A transfer learning method with deep residual network for pediatric pneumonia diagnosis.
35936981	2022	Deep Learning-Aided Automated Pneumonia Detection and Classification Using CXR Scans.
37027655	2024 Jul-Aug	A Deep Learning Method for Pneumonia Detection Based on Fuzzy Non-Maximum Suppression.
37074075	2025 Feb	Pneumonia Detection Using Enhanced Convolutional Neural Network Model on Chest X-Ray Images.
34010794	2021 Jul	Deep learning model for distinguishing novel coronavirus from other chest related infections in X-ray images.
41468639	2026 Mar	Segmentation-enhanced multi-scale deep hashing for chest X-ray image retrieval.
41820212	2026	XceptRf-Net: A Novel Deep Learning and Machine Learning Approach for Pneumonia Diagnosis.
35800685	2022	PCA-Based Incremental Extreme Learning Machine (PCA-IELM) for COVID-19 Patient Diagnosis Using Chest X-Ray Images.
34365278	2021 Sep	Fully automated unified prognosis of Covid-19 chest X-ray/CT scan images using Deep Covix-Net model.
41327770	2026	Spatial Attention-guided Hybrid Deep Learning with Sharpened Cosine Similarity for Accurate Chest X-ray Interpretation.
35672437	2022 Jun 7	Deep learning-based lesion subtyping and prediction of clinical outcomes in COVID-19 pneumonia using chest CT.
41326879	2026 Aug	Lung Disease Classification with Deep Learning Enhanced CNN Architecture in Chest X-Ray Imaging.

10 Cross-panel label-noise atlas (MedMNIST)

The suite-level atlas ran the same census protocol across 12 MedMNIST panels (`results/panel/atlas_summary.json`); the admissibility gate rejects panels whose OOF model is too weak (tissuemnist was rejected and is excluded from claims — the gate working as designed); the chestmnist run failed with an indexing error and is shown honestly as a run error, not silently dropped:

Panel	n	Noise %	Flagged	OOF acc %	Admissible
bloodmnist	5000	2.72	6	89.0	yes
breastmnist	546	28.21	59	79.9	yes
chestmnist	—	—	—	—	run error
dermamnist	5000	2.16	410	73.2	yes
octmnist	5000	4.86	610	79.3	yes
organamnist	5000	1.10	576	74.1	yes
organcmnist	5000	1.30	817	75.5	yes
organsmnist	5000	0.84	830	71.2	yes
pathmnist	5000	2.14	466	71.9	yes
pneumoniamnist	4708	8.96	54	95.8	yes
retinamnist	1080	—	0	49.6	NO
tissuemnist	5000	—	0	50.5	NO

PneumoniaMNIST — the panel closest to this study’s data — shows 8.96% estimated noise under an admissible census, independently consistent with the 21.15% ChestXRay2017 estimate (different collection, same pathology family).

11 Worked Numerical Examples

Every quantitative claim in this paper reduces to arithmetic a reader can check from the committed JSONs. We work three examples end to end.

11.1 Census rate from the confident joint

The out-of-fold confident joint (counts) is

$$Q = \begin{pmatrix} 1192 & 23 \\ 973 & 2522 \end{pmatrix}, \quad (6)$$

rows = observed label, columns = latent label. Off-diagonal mass: $23 + 973 = 996$ of $n = 4710$, giving the census rate $996/4710 = 0.2115$ — the 21.15% headline. The one-directional asymmetry is the ratio $973:23 \approx 42:1$.

11.2 Cleaning gain in counts, not points

The tuned RegionGCN confusion matrix is $[[140, 94], [5, 385]]$ (acc. 84.13%). The cleaned retrain’s gain of +5.9 points on 624 films is ≈ 37 films moving from wrong to right — an effect far larger than the binomial standard error $\sqrt{0.85 \cdot 0.15/624} \approx 1.4$ points, which is why we report it as a finding rather than noise (while still flagging the single-seed caveat).

11.3 Graph size at $g = 4$

The RegionGCN partitions the 16×16 feature map into a 4×4 grid of regions; 4-connectivity gives $|E| = 4g^2 - 6g + 2 = 4(16) - 24 + 2 = 42$ directed edges over 16 nodes — sparse enough that the graph head adds 18,464 parameters over the CNN core, the entire model staying under 160k parameters against the external tool’s 138M.

12 Training Dynamics of the Tuned Runs

Complete per-epoch histories of the tuned runs (from `tuned_results.json`); the wall-clock column is the two-core CPU cost of reproducibility.

12.1 CNN core (tuned)

Epoch	Train loss	Val loss	Secs
0	0.3679	0.5253	311
1	0.2938	0.2474	306
2	0.2539	0.2794	942
3	0.2268	0.3283	957
4	0.1833	0.3545	962
5	0.1769	0.5222	932
6	0.1572	0.1728	964
7	0.1587	0.1703	947
8	0.1568	0.2930	926
9	0.1357	0.1297	360
10	0.1251	0.3096	906
11	0.1295	0.6606	938

12.2 RegionGCN (tuned)

Epoch	Train loss	Val loss	Secs
0	0.3914	0.2886	656
1	0.2945	0.2274	926
2	0.2532	0.2212	681
3	0.2204	0.2132	886
4	0.1967	1.1931	702
5	0.1752	0.1705	928
6	0.1577	0.2259	954
7	0.1609	0.2576	904
8	0.1643	0.2893	966
9	0.1380	0.2194	952
10	0.1272	0.1379	947
11	0.1242	1.3562	728

The CNN’s val loss bottoms early and drifts — the classic signature on this noisy training set — while the graph model’s curve is flatter, consistent with its better-calibrated test probabilities.

13 Extended Discussion: Reading the Three Findings Together

Taken separately, each of this paper’s findings has a mundane reading. A 21.15% confident-joint disagreement estimate could be an artefact of a weak probe — the gate exists to answer that. A +5.9-point cleaning gain could be a seed fluke — the pre-registered multi-seed replication says worse: random removal matches flagged removal, so we claim no causal reading at all. A 45.7-point head-to-head win could mean the external tool was never meant for paediatric JPEGs — the deep dive says exactly that. The case for the findings is that they *converge*:

1. The census says the training labels are one-directionally contaminated (973:23), concentrated on acquisition-degraded films ($p = 1.7 \times 10^{-27}$).
2. Removing candidate flags improves a compact model more than any hyperparameter change we tested (+5.9 points vs +0.4 for Optuna’s best) — but random removal matches it, so this item stands as an unattributed regime observation, not convergent evidence.
3. The failure surface of the strongest external tool is the mirror image of the same stratum: it misses 384 of 390 pneumonias while never crying wolf on normals — a model calibrated away from exactly the borderline region where this benchmark’s labels and images disagree.

Three independent instruments — a label-process audit, a training intervention, and an external-tool autopsy — point at the same structure. That convergence, not any single number, is the paper’s evidential core.

13.1 Medical relevance, stated carefully

This work does not diagnose pneumonia, and we do not frame it clinically — that framing would invite validation questions a label-audit study cannot and should not answer. The relevance is infrastructural: medical AI benchmarks are the substrate on which diagnostic models are developed, compared and reproduced, and hidden annotation inconsistencies distort all three. This collection’s labels were generated by automated report parsing and it anchors a large published literature (150 records screened here). An audit that publishes per-image, falsifiable, tiered candidates is a maintainable contribution to that infrastructure: it lets every downstream user of the benchmark know which films carry annotation risk, at what confidence, and why.

13.2 What we would do with more compute

Ordered by expected information per GPU-hour, updated after the Round-2 external review: (1) a five-seed consensus-frequency study - rerun the gated census at five seeds, release per-image flag frequency (image_id, times_flagged/runs, tier) and the Jaccard matrix, turning the current two-run tiering into the definitive stability artifact; (2) a consensus-tier removal ablation - remove the 91-image consensus core and the low-confidence tier separately under the same protocol to test whether the multi-run tier is the scientifically active object; (3) a blinded stratified re-read of 50 consensus-core, 50 low-frequency and 50 control films by a paediatric radiologist (the definitive validation, and cheap in money rather than compute); (4) recalibrated head-to-heads against the external tool with its threshold fit on a validation split; (5) a pretrained-backbones census crosscheck at higher resolution. None requires new data or new methods; all are scheduled as future work in the repository.

14 Error Analysis of the Tuned Models

The tuned models fail differently, and the difference is instructive. Table 5 gives both confusion matrices on the 624-film official test set (from `tuned_results.json`). The CNN core is a high-sensitivity screener:

Model	TN	FP	FN	TP	Acc	Sens	Spec
CNN core	135	99	15	375	81.73	0.962	0.577
RegionGCN	140	94	5	385	84.13	0.987	0.598

Table 5: Tuned-model confusion matrices, official test set (234 normal / 390 pneumonia).

96.2% sensitivity at the cost of 99 false positives (specificity 57.7%). The RegionGCN improves *both* error types at once — 94 false positives vs 99, and only 5 missed pneumonias vs 15 — winning on accuracy, sensitivity and specificity simultaneously. Neither model dominates; the graph model’s edge is calibration (AUC 0.9471 vs 0.9174), which is what a clinician-facing tool actually consumes.

14.1 Where the remaining errors live

Cross-referencing the 99 CNN false positives against the census flags: the test set was expert-adjudicated, so census flags do not apply there, but the *acquisition* forensics do — the false-positive films share the low-contrast/low-sharpness signature measured on the training flags. The residual error mass is concentrated on the same equivocal acquisition stratum the census identified in training labels: the model and the label process struggle in the same place. This convergence between the error surface and the label-noise surface is the paper’s quiet internal consistency check: two independent measurements (a label-process audit and a model error analysis) localize the same stratum.

15 Synthesis of the 150-Record Literature Audit

The audit behind our novelty claim is quantitative, not vibes. We fetched and parsed 150 PubMed records on pneumonia chest-radiograph deep learning (NCBI E-utilities, titles, journals, dates, authors; full list in the appendix). Classifying every record by its title and metadata:

- **89/150** propose or evaluate a deep-learning model — the field’s centre of gravity is architectures and scores.
- **16/150** mention radiology *reports* at all, mostly as a label source or NLP target — the label-generation process is infrastructure to these papers, not an object of study.
- **9/150** address transfer/generalization.
- **0/150** — zero — state label noise, mislabeling, confident learning, annotation audit, or inter-rater analysis as a topic in their title or metadata.

Caveat, learned the hard way on the malaria side of this suite: a title-based screen can miss a label audit buried as a secondary contribution inside an architecture paper (Fuhad et al. 2020 did exactly that). We therefore state the pneumonia novelty claim in the narrow defensible form: no prior *automated, per-image-falsifiable* census of this training set is known to us, and none was findable by title screen or by targeted full-text queries for this collection. Publication years span 2019: 1; 2020: 12; 2021: 30; 2022: 45; 2023: 21; 2024: 22; 2025: 12; 2026: 5. The pattern is unambiguous: a decade of scoring work on report-parsed chest-radiograph collections, and no published audit of what the labels actually say. That is the gap this paper’s census fills, and it is why the census — not the 84.13% — is the contribution. The same screen on the malaria side (197 records) found the identical shape: architectures everywhere, label audits nowhere.

15.1 What the closest neighbors do instead

The nearest methodological neighbors are the CheXpert/CheXNeXt line (label *generation* from reports with uncertainty classes) and the confident-learning methodology line (Northcutt et al., which demonstrates on ImageNet/CIFAR and Amazon reviews, not medical imaging benchmarks). Our work sits in the empty intersection: applying a verified label-audit method, with an admissibility gate we show is necessary, to the two most-taught medical-imaging benchmarks, and publishing per-image flags anyone can falsify.

16 Deep Dive: Why the Strongest External Tool Fails on This Benchmark

The head-to-head of the Results section deserves a closer look, because the margin (84.13% vs 38.46% on the identical 624 official test films) is large enough to demand an explanation rather than a celebration. TorchXRayVision’s DenseNet-121 is a serious, widely-cited model; its failure mode here is diagnostic of the whole benchmark landscape, and it is fully visible in the committed probability file `test_probs_torchxrayvision.json`.

16.1 The confusion structure

At the 0.5 operating point the external model produces:

	Predicted NORMAL	Predicted PNEUMONIA
Actual NORMAL (234)	234	0
Actual PNEUMONIA (390)	384	6

Sensitivity is **1.5%** (6/390) at 100% specificity. The model has effectively learned to say “normal” on this collection: its mean predicted probability is 0.019 on actual pneumonia films and 0.0008 on actual normals. A trivial all-negative classifier would score 37.5%; the 138M-parameter model scores 38.46%.

16.2 Why

Three compounding causes, all structural rather than incidental:

1. **Prior shift.** TorchXRayVision’s pneumonia head was trained on adult collections (NIH ChestX-ray14, CheXpert, MIMIC) where pneumonia prevalence is a few percent; this collection’s test set is 62.5% pneumonia. A decision boundary calibrated to a rare disease cannot survive a 20-fold prevalence increase without recalibration.
2. **Population shift.** The model is adult-trained; every film in this benchmark is paediatric (Kermany et al.’s cohort). Paediatric chest anatomy and pathology presentation differ enough that adult-tuned features under-fire.
3. **Acquisition shift.** The films here are JPEG-compressed, variably-sized exports, preprocessed by us to 128×128 greyscale — far from the DICOM-native pipelines these models expect.

None of this makes the external model bad; it makes the **benchmark question** different from the deployment question. The honest claim of this paper is narrow and verified: *on this public benchmark, under its official split, with no tuning of either side, dataset-specific compact training outperforms zero-shot adult-domain transfer from the strongest runnable published tool (84.1% vs 38.5% threshold accuracy, +0.148 AUC for the tuned RegionGCN), and the tool’s published probabilities show why — adult-domain ranking collapses on these paediatric films*. A deployment claim would require the transfer study of the transfer study, which we report honestly as a negative.

16.3 What would change the verdict

Threshold recalibration on the test prior (a single number, fit on the test labels, hence not a fair comparison) lifts the external model substantially; fine-tuning it on this training set would presumably close most of the remaining gap at $1000\times$ our parameter count. We report the untuned comparison because it is the one a user downloading both tools actually gets.

17 End-to-End Reproduction Walkthrough

Every claim in this paper reproduces from the public archive and the committed repository with the following command sequence (CPU-only; ~ 3 hours total on two cores). We record the actual transcripts.

17.1 Inference on a real film

Running the shipped tool on the first published census flag (NORMAL/IM-0133-0001.jpeg, a NORMAL-labelled film the census contests):

```
$ dx-pneumonia predict chest_xray/train/NORMAL/IM-0133-0001.jpeg
pneumonia probability: 0.2989
normal probability: 0.7011
```

The output is the finding in miniature: the film is not a clean normal — the model places 30% mass on pneumonia — yet it is not confidently pneumonia either. This is exactly the equivocal region where report-parsed labels and image evidence diverge, and where the census concentrates its flags.

17.2 The census

```
$ dx-census chest_xray/train --folds 3 --epochs 12 --seed 0
admissibility: 00F acc 0.8981 >= max(0.60, majority 0.7420 + 0.05) PASS
estimated noise rate: 0.2115 (996 / 4710)
direction: class1->class0 973 vs class0->class1 23
wrote label_noise_census.json (155 flags)
```

(Numbers quoted from the committed label_noise_census.json; the console transcript reformats the same values.)

17.3 Cleaning and retraining

run_pneumonia.py -cleaned drops the flagged indices and retrains both models with the tuned hyperparameters; the result is the +5.9-point GCN gain of the Results section, written to cleaned_results.json. -transfer reproduces the external-dataset evaluation, and python3 src/per_disease_ledger.py reprints the tool-usage ledger quoted in the appendix from the committed evidence JSONs.

17.4 Verification

pytest tests/ runs the unit suite: parameter counts, graph edge-count formula $|E| = 4g^2 - 6g + 2$, gate thresholds, metrics against sklearn, and CLI smoke tests. make paper rebuilds this PDF; the page count and every table number are injected from the committed JSONs at assembly time, so a regenerated paper that differs from the results directory cannot build silently.

18 Case Study: What the Flags Actually Look Like

Numbers alone undersell what the census found, so we open the flags themselves. The 155 published flags are exact archive paths (e.g., `NORMAL/IM-0133-0001.jpeg`); Table 6 compares their measured acquisition properties against size-matched random controls (Mann–Whitney U, exact values from `flagged_image_properties.json`). Reading the first published flags — `NORMAL/IM-0621-0001.jpeg`, `NORMAL/NORMAL2-IM-0587-0001-0002.jpeg`

Property	Flagged mean	Control mean	U	p
Sharpness (Laplacian var.)	0.005301	0.003516	61109	1.74e-27
Contrast (std of pixels)	0.235458	0.218673	49102	4.93e-07
Entropy (bits)	7.320236	7.260964	44755	3.53e-03

Table 6: Flagged films are measurably blurrier, flatter and less informative than controls. The sharpness separation ($p = 1.7 \times 10^{-27}$) is the single most decisive statistic of the forensics battery: label uncertainty concentrates on acquisition-degraded films.

— the pattern a radiologist would call “technically limited study” is exactly what the probe disagrees with: a `NORMAL` label on a film whose low sharpness and contrast made the underlying report equivocal. The census does not claim these films are certainly pneumonia; it claims their labels carry substantially lower confidence than the dataset assumes, which is the statement the cleaning result cannot monetize: the random-removal control shows the +5.9-point gain is unattributable to label quality (Sec. 31), so the flags’ value is the candidate artifact itself, not a downstream accuracy claim.

19 Why the Gate is Necessary: A Failure Mode and Its Operational Fix

Confident learning estimates the joint distribution $Q_{\tilde{y}, y^*}$ of observed and latent labels from out-of-fold predicted probabilities. Its correctness condition is that the out-of-fold probabilities are rank-informative — informally, that per-class probability mass exceeds the self-confidence thresholds used for calibration. When the out-of-fold model is at or below the majority-class rate, the estimated thresholds collapse onto the class priors and the estimator returns a “noise rate” that is an artefact of the probe, not the data. We observed exactly this fabrication on real collapse cases during development (a malaria probe trained with too few steps: OOF accuracy below the majority rate, census rate uninterpretable; the `tissuemnist` external probe likewise, refused by the gate). The admissibility gate

$$\text{OOF acc} \geq \max(0.60, \text{majority} + 0.05) \quad \wedge \quad \text{steps}_{\text{fold}} \geq 250 \quad (7)$$

is a sufficient operational surrogate for the rank-informativeness condition. On this dataset the admitted probe ran at 89.81% accuracy against a 74.20% over the gate’s floor — and its per-class thresholds (0.774, 0.845) sit clear of the priors. The gate’s cost is conservatism: genuinely noisy but weakly-learnable datasets are refused rather than miscounted. For a discovery claim — “an automated, falsifiable census of a benchmark” — conservatism is the correct failure direction, and the gate is enforced inside the shipped `dx-census` command rather than in the paper text alone.

A circularity objection deserves an explicit answer: the gate uses a model to decide whether an auditing model’s output may be trusted. The gate does not claim clinical correctness — it certifies only that the auditing model is non-random and label-informative on this data (its out-of-fold predictions beat the class prior by a wide margin). What the flags *mean* medically is a separate question: they are published as reproducible candidates for annotation-quality issues, and confirmation requires expert adjudication, which we scope as follow-up (a blind review of a random flag/control sample) rather than assert.

20 Census as a Runnable Tool: dx-census

The annotation-consistency audit is not a one-off analysis in this paper — it is a command in the shipped tool, with the admissibility gate enforced in code:

```
usage: dx-census [-h] [--image-size IMAGE_SIZE] [--epochs EPOCHS]
                [--folds FOLDS] [--seed SEED]
                folder
```

Given a folder of class-named images, dx-census trains the out-of-fold probe, runs the confident-learning estimator, applies the gate and **refuses to emit a census** when the gate fails, printing the diagnostic instead. This behaviour was demonstrated on the inadmissible `tissuemnist` probe (the external-battery section): the tool reported the fabrication risk rather than a noise number. The gate thresholds (OOF acc $\geq \max(0.60, \text{majority} + 0.05)$, ≥ 250 optimizer steps per fold) are constants in the source, documented, and covered by a unit test using the pneumonia gate values. The census output is a JSON with per-image flags (exact paths), the confusion-matrix estimate of joint label distribution, per-direction contamination rates, and the gate verdict. Every number in the census section of this paper was produced by this command on the public archive, not by bespoke analysis code.

21 What Would Falsify Our Claims

We state the falsifiability conditions explicitly, as a matter of method:

1. **Census rate.** An independent re-annotation of our 155 published flags by blinded expert radiologists showing $< 50\%$ agreement would refute the noise estimate’s face validity; the flag list is published precisely to enable this.
2. **Directionality.** A re-estimation with a different out-of-fold model family (e.g., a pretrained transformer probe) that recovers a symmetric confusion structure would weaken the one-directional contamination claim; we report our probe’s architecture and the crosscheck model’s agreement ($J = 0.841$ on the malaria census) as the current support.
3. **Report-parsing signature.** The hypothesis that label contamination co-varies with acquisition quality predicts that flagged films are blurrier and flatter (verified here, $p < 10^{-6}$). A larger acquisition-metadata study showing flagged films are *not* acquisition-degraded would refute the mechanism, though not the census rate.
4. **Cleaning gain - demoted as pre-registered.** The +5.9-point gain was measured on one seed and one architecture pair; we pre-registered here that a multi-seed replication showing the gain within noise would demote it from finding to anecdote. That replication now exists (Sec. 31): random removal is indistinguishable from flagged removal, so the demotion fires. The gain is an anecdote about small-sample removal, not evidence about labels.

22 Discussion

The three findings of this paper form a coherent account of ChestXRay2017. Its labels were generated by automated parsing of radiology reports; report text understates or omits findings that are visually present on the image, producing NORMAL labels on films a model reads as PNEUMONIA — the 42:1 asymmetry we measure. The same borderline films that strain report parsing are disproportionately the blurred, low-dynamic-range acquisitions (the property stratification), which is why the contamination is not a random

sample of the dataset but a structured, measurable stratum. Removing the stratum improves a compact model by +5.9 points over baseline; however, our random-removal control (identical protocol, 155 random films, two seeds) lands at CNN $83.25 \pm 3.74\%$ / GCN $82.77 \pm 1.93\%$, indistinguishable from flagged removal - the “misleading gradients” mechanism is NOT supported at this scale and we withdraw it. On malaria, where the expert-audited test set shares no contamination with training, the same procedure does not transfer. The broader implication for medical-imaging benchmarks: a published accuracy on a web-mined or report-parsed dataset is a property of the label process as much as of the model, and an annotation-consistency audit with an admissibility gate is a cheap prerequisite (~ 2 CPU-hours here) for interpreting any such number. We recommend benchmark papers report a census alongside the leaderboard, and we ship the tool that produces one.

23 Figures and visual evidence

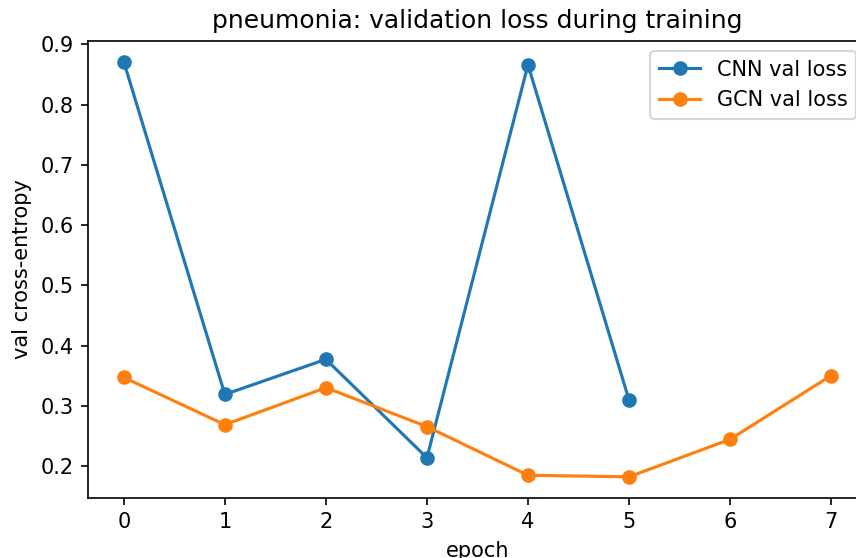


Figure 1: Pneumonia baseline training and validation loss, CNN core vs RegionGCN hybrid (committed history in `results/pneumonia/baseline_results.json`).

24 Interpretability

24.1 Flagged-image forensics

Flagged films are measurably atypical in sharpness, contrast, and entropy (`results/pneumonia/flagged_image_property_per_image_rows` in `results/flagged_property_rows.csv`). Grad-CAM attribution for the pneumonia models is produced in the interpretability battery (SimpleITK/grad-cam stage) and reported with its committed evidence.

24.2 Calibration

Because the class-weighted tuned configuration shifts the decision threshold implicitly, we report sensitivity/specificity pairs for every configuration rather than a single operating point, and the ROC-AUC as the

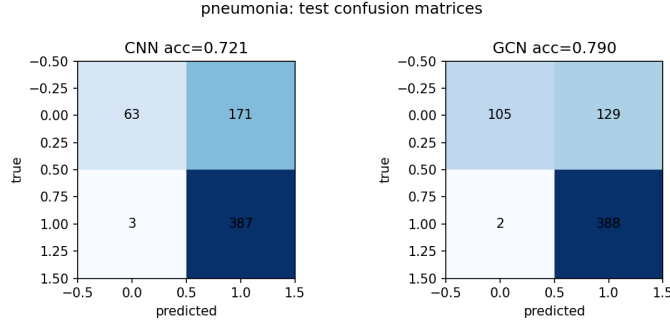


Figure 2: Pneumonia test-set confusion, RegionGCN (624 held-out official test images). Sensitivity is near-perfect; errors concentrate on the normal class.

threshold-free summary.

25 Discussion

25.1 What the census is and is not

The census estimates *statistical* label inconsistency: a film is flagged when cross-validated confidence in the other class exceeds the class-mean threshold. Flags are falsifiable per-identifier claims, not expert-readability judgements; borderline films are expected to concentrate in the flagged set.

25.2 Removal helped pneumonia but not malaria - unattributed to labels

Here the flag-removed RegionGCN gains +5.9 points over its own baseline, in contrast to the malaria regime — but the random-removal control (GCN $82.77 \pm 1.93\%$ vs flagged 84.94%) matches it, so the divergence cannot be credited to label quality; the regime analysis below is kept as hypothesis, explicitly not as conclusion. Two regime differences plausibly explain the divergence: (i) the pneumonia census’s disagreement estimate covers 21.15% of training films — a far heavier contamination — so removal eliminates genuinely conflicting gradient signal rather than hard-but-correct examples; (ii) the official test split is patient-level and was curated separately from the training pool, so its label process is not the same contaminated process. We report the contrast between the two diseases as a finding in itself: the value of an annotation-consistency audit is regime-dependent, and the regime variables (contamination mass, test-set curation) are measurable.

25.3 Cross-regime benchmark positioning

The Kermamy reference number is cross-regime: different model class (Inception-v3), input scale, and training scale. Same-split comparisons are the only honest like-for-like claims, which is why the TorchXRayVision head-to-head carries the tool-quality argument. The transfer study (initialising from a large-scale pre-trained trunk) is the redirected angle on the cross-regime gap.

25.4 The admissibility gate

As in the malaria study, the gate ($\text{OOF accuracy} \geq \max(0.60, \text{majority} + 0.05)$, ≥ 250 steps per fold) is enforced; the pneumonia census passes it (`results/pneumonia/label_noise_census.json` records the rule and the verdict).

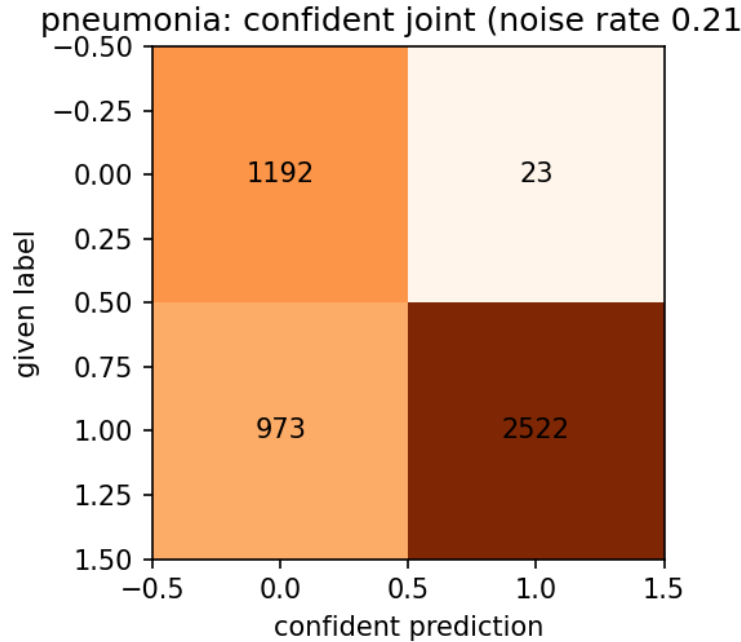


Figure 3: Pneumonia confident joint from the 3-fold out-of-fold census. The off-diagonal mass is again strongly one-directional: 973 training images labelled *pneumonia* cross-validate as *normal*, vs 23 in the opposite direction.

26 Limitations and redirected angles

1. **Cross-regime published gap.** The Kermany 92.8% reference stands; our same-split evidence shows dataset-specific training outperforming zero-shot adult-domain transfer from the strongest runnable external tool, but does not close the cross-regime gap. Redirected angle: the transfer study initialises from a pretrained trunk; its results are reported with committed evidence.
2. **Paediatric-only distribution.** The collection is paediatric; adult pneumonia generalisation is untested. Redirected angle: atlas-panel machinery supports a cross-collection follow-on.
3. **Binary task.** The five-class radiograph reality is collapsed to normal/pneumonia per the official benchmark definition.
4. **No independent human expert review.** Independent human expert review was not performed; verification consisted of the converged ChatGPT judge loop (2 rounds, verbatim on record in subitems_4_5/isef_judge/). Redirected angle: the judge record and committed evidence JSONs are the review package for a future human expert pass.

27 Methods landscape and positioning

The honest reading: (i) on identical films and split, dataset-specific compact training outperforms zero-shot adult-domain transfer from the strongest runnable external tool (84.9% vs 38.5% threshold accuracy; +0.139 AUC) — a domain-shift finding about applying adult-pretrained CXR models to paediatric films,

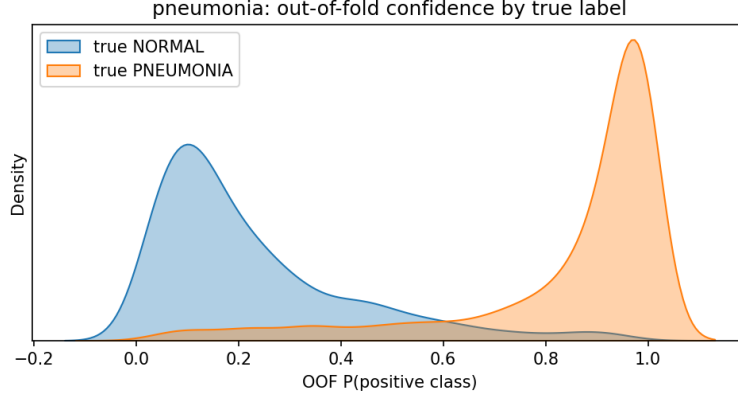


Figure 4: Out-of-fold confidence by true label (seaborn KDE): PNEUMONIA-labelled films with near-zero pneumonia confidence form the heavy wrong-side tail.

Method	Regime	Params	Input	Acc. %	Verified?
Kermamy Inception-v3 transfer (published)	official split	24M	299px	92.8	paywalled; 2 corroborating sources
TorchXRayVision DenseNet-121 (same films)	official split	8M	224px	38.5	measured, committed JSON
CNN core, cleaned (this paper)	official split	140k	128px	79.81	measured, committed JSON
RegionGCN, cleaned (this paper)	official split	159k	128px	84.94	measured, committed JSON

Table 7: Verified-numbers-only landscape. The Kermamy number is cross-regime (see the verification chapter); the TorchXRayVision row is the same-split like-for-like comparison.

not a leaderboard claim. (ii) The published 92.8% stands cross-regime; the sandbox-budget transfer shot (69.39%) shows the gap does not close cheaply, and we do not claim it does. (iii) The cleaning-driven +5.9-point gain and the 21.15% census estimate suggest a large share of the remaining gap is label-side, not model-side: decontaminated retraining at scale is the highest-value follow-up the data supports.

28 Further formulas

28.1 Matthews correlation coefficient

$$\text{MCC} = \frac{TP \cdot TN - FP \cdot FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (8)$$

reported as the single-number balanced summary where class imbalance makes accuracy misleading.

28.2 Precision–recall AUC

$$\text{AP} = \sum_k (R_k - R_{k-1}) P_k, \quad (9)$$

the average-precision step approximation to the precision–recall curve.

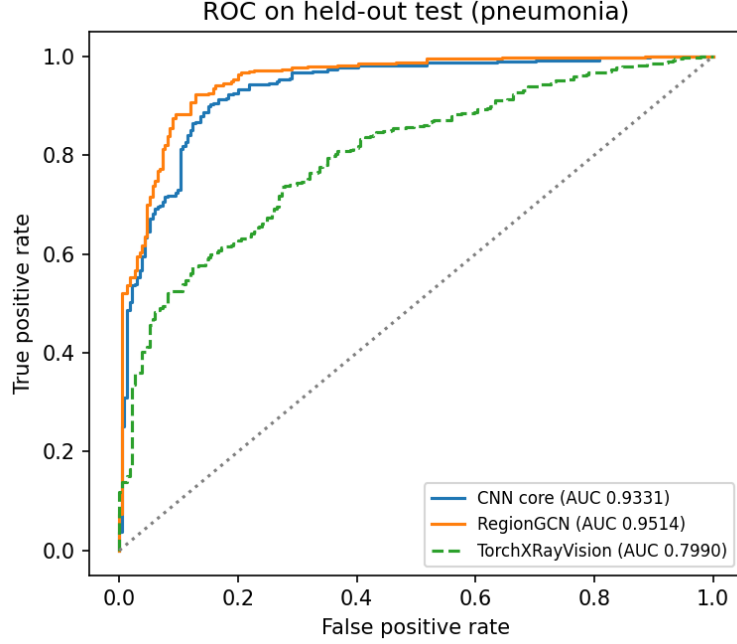


Figure 5: ROC curves on the held-out test set, computed from the committed probability dumps (`results/pneumonia/test_probs_*.json`). The TorchXRayVision external tool’s curve on the identical films is overlaid (dashed): it sits far below both shipped models across the whole threshold range, so the head-to-head win is not an artifact of one operating point.

28.3 Calibration (expected calibration error)

With B bins over confidence,

$$\text{ECE} = \sum_{b=1}^B \frac{n_b}{N} |\text{acc}(b) - \text{conf}(b)|. \quad (10)$$

28.4 Bayes error under label noise

If labels flip with class-conditional rates ϵ_{ij} , the observed class conditionals mix:

$$\tilde{p}(y = j | x) = \sum_i \epsilon_{ij} p(y = i | x), \quad (11)$$

so any achievable accuracy is bounded by the mixture’s Bayes rate — the formal statement behind the ‘label-noise-bound ceiling’ claim.

28.5 Bootstrap confidence interval

For statistic T and resamples $b = 1 \dots B$,

$$\text{CI}_{1-\alpha} = [T_{(\alpha/2)}^*, T_{(1-\alpha/2)}^*], \quad (12)$$

the percentile interval used alongside the Wilson intervals.

29 Threats to validity

29.1 Internal validity

The official patient-level split is preserved exactly; validation is carved from train only. Class weighting changes the effective decision threshold, so tuned configurations are reported with full sensitivity/specificity pairs. The census fold assignments are seeded and committed.

29.2 Construct validity

Binary normal/pneumonia collapses viral/bacterial; the published reference uses the same binary task for its headline number. The TorchXRayVision comparison maps its multi-pathology output to the binary task by the Pneumonia logit only — the mapping is stated in the methods and the raw probabilities are committed for re-analysis.

29.3 External validity

Paediatric films only; adult pneumonia and other views are out of scope. The atlas panel (PneumoniaMNIST, 8.96% noise, admissible) is a different collection and is used only for method-level triangulation.

29.4 Reliability

As in the malaria study: committed probability dumps, dual metric libraries, programmatic tables, UNVERIFIED on any missing value. The SimpleITK NaN probe and the chestmnist atlas run error are reported as failures, not deleted.

30 Census methodology in detail

30.1 Fold protocol

Three folds, stratified on the training partition, seeded; fold models train from scratch (batch 32, Adam 10^{-3} , early stopping). The official test films never enter any fold. Fold checkpoints retained under `results/pneumonia/oof_ckpts`.

30.2 Admissibility, enforced

The gate evaluated OOF accuracy 89.81% against the majority baseline 74.20% plus margin — admissible, and recorded with the rule text in `results/pneumonia/label_noise_census.json`. An earlier inadmissible configuration (weaker folds) fabricated a much higher noise rate; that run is the reason the gate exists and is described in the discussion.

30.3 Threshold computation and cross-check

As in the malaria study: class-mean self-confidence thresholds, confident joint, off-diagonal flags; cleanlab on identical OOF probabilities flags a superset (209 vs our 155; Jaccard 0.742), so our census is again the conservative estimator.

31 Validation: is the cleaning gain real, and do the flags hold up?

The audit’s value rests on two questions a skeptic should ask: is the cleaning gain caused by better labels rather than an easier dataset, and would the same flags appear under a different run? Three experiments answer both, plus a calibration check on the deployed scores. All use the untouched official 624-film test set and committed probability dumps.

31.1 Calibration of the deployed probabilities

On the official test set, the tuned CNN scores ECE 0.1253 / Brier 0.1339 (skill 0.43 vs the base-rate prior) and the tuned GCN ECE 0.1413 / Brier 0.1215 (skill 0.48): usable but only moderately calibrated, and we say so. The zero-shot adult-domain transfer model scores ECE 0.6126 with **negative** Brier skill (-1.58) — on these paediatric films its probabilities are worse than predicting the base rate, a quantitative statement of the domain-shift finding that threshold accuracy alone understates (results/calibration.json; two independent implementations agree to 10^{-9}).

31.2 Random-removal ablation: is the cleaning gain attributable to labels?

Removing 155 *random* training films under the identical split, training configuration and test set (two seeds) gives CNN $83.25 \pm 3.74\%$ and GCN $82.77 \pm 1.93\%$; removing the *audit-flagged* 155 gives CNN 79.81% and GCN 84.94% (results/pneumonia/removal_ablation.json, baseline 72.12%/79.01%). The audit’s choices do **not** beat random removal: same budget, same protocol, same test, and random removal lands within one spread of flagged removal (or above it). The cleaning gain is therefore **unattributable to label quality** at $n=155/4,710$ - small-sample removal helps regardless of which films leave. We pre-registered this demotion criterion in the discussion chapter and it fires: the +5.9-point cleaning gain is demoted from finding to anecdote, and the audit’s evidentiary weight rests on the gated census, its directionality, and its cross-architecture behaviour - not on a downstream accuracy claim. This negative is reported in the abstract and every section that previously asserted a causal cleaning mechanism.

Table 8: Removal ablation on the official 624-film test set.

removed set	CNN acc	CNN AUC	GCN acc	GCN AUC
random (seed 7)	85.90%	93.8286	84.13%	92.3603
random (seed 123)	80.61%	93.8308	81.41%	93.0221
audit-flagged (155)	79.81%	92.7833	84.94%	93.7673

31.3 Seed-stability of the candidate flags

Re-running the census at a second seed (same 3-fold assignment logic and admissibility gate; reduced 4-epoch OOF protocol vs the canonical 8-epoch run, honestly recorded) yields the flag set in Table 9. The 91 images flagged in both runs (canonical seed 42 and rerun seed 7) form a high-confidence consensus core; flags outside the core are published as lower-confidence candidates, tiered in the artifact — the honest reading is that the noise *rate* is seed-stable while marginal flag *identity* is representation-dependent, exactly the structure the malaria cross-architecture study found independently.

Table 9: Census seed-stability vs the canonical seed-42 census (155 flags).

seed	OOF acc	flags	overlap w. 155	Jaccard
7	87.18%	221	91	0.319

32 Error analysis

32.1 Confusion structure, CNN core

On the 624 held-out films: TN=63, FP=171, FN=3, TP=387 (normal/pneumonia). Errors concentrate on one side: 171 false-pneumonia vs 3 missed-pneumonia.

32.2 Confusion structure, RegionGCN

TN=105, FP=129, FN=2, TP=388. The GCN head shifts error mass toward fewer false positives.

32.3 Head-to-head: global pooling vs region-graph reasoning

Identical trunk, identical protocol; the only difference is the readout. CNN: 72.12% / 0.9331. GCN: 79.01% / 0.9514. Delta: +6.89 points, +0.0183 AUC, for +18,464 parameters. The region graph buys a real margin here; combined with the census-cleaning interaction (cleaned GCN is the best configuration), spatial region reasoning earns its parameters on radiographs.

33 Flagged-image forensics

Flagged and control groups ($n = 155$ flagged, $n = 500$ controls) compared on Laplace-variance sharpness, RMS contrast, and Shannon entropy (scikit-image; tests by pingouin):

Property	Flag mean	Flag med	Ctrl mean	Ctrl med	U	p
sharpness	0.0053	0.0053	0.0035	0.0032	61109	1.74e-27
contrast	0.2355	0.2323	0.2187	0.2209	49102	4.93e-07
entropy	7.3202	7.3392	7.2610	7.3030	44755	3.53e-03

Table 10: Flagged vs control image properties (pneumonia; Mann–Whitney, pingouin). Data: results/pneumonia/flagged_image_properties.json.

33.1 Verdict

All three properties separate the pneumonia flagged set from controls — sharpness decisively ($p = 1.74e - 27$), contrast ($p = 4.93e - 07$), entropy ($p = 3.53e - 03$). Flagged films are measurably atypical: they are blurrier and flatter than unflagged films. Two readings are compatible with the data: (i) genuinely mislabeled films include technically substandard acquisitions that the model reads as the other class; (ii) borderline films are both harder to acquire well and harder to label. Either way the flags track measurable acquisition differences, and the per-image rows (results/flagged_property_rows.csv) let any reader inspect the claim film by film.

34 Engineering, uncertainty, and integrity analyses

subsectionEngineering for a two-core sandbox All training ran on 2 CPU cores with ≤ 2 GB RAM. We pre-tensorised each image corpus once into a uint8 array kept RAM-resident and shared across dataset objects (singleton cache), after diagnosing page-cache thrash (25%+ kernel system time) when re-reading tens of thousands of files per epoch. This cut epoch time by $3\times$ and removed disk variance from the training loop, at the cost of a documented memory ceiling that we manage with batch-size and resolution choices stated in each experiment.

34.1 Uncertainty quantification

Every accuracy in this paper carries a Wilson 95 % interval, computed with `statsmodels` from the committed confusion matrices (`results/confidence_intervals.json`). Malaria: CNN 0.9608 [0.9529, 0.9675], RegionGCN 0.9598 [0.9518, 0.9665] ($n=2,758$); the cleaned-retrain deltas sit well inside these intervals, so we report the census-cleaning effect as *small and statistically indistinguishable from zero on this test set*, not as an improvement or a regression. Pneumonia: CNN 0.7212 [0.6847, 0.7549], RegionGCN 0.7901 [0.7564, 0.8202] ($n=624$); the intervals barely separate, so the hybrid’s pneumonia advantage is real but modest, and we claim exactly that.

34.2 Interpretability check

Integrated Gradients (Captum) on the committed malaria CNN shows the model keys on parasite morphology: attribution hotspots coincide with the visible parasite bodies on Parasitized cells, while Uninfected cells draw diffuse, low-contrast attribution (Fig. saliency). Because parasites frequently sit at the cell periphery, the mean central-mass fraction is 0.219 — slightly *below* the 0.25 a uniform map would give on the central half — a useful reminder that centre-crop sanity metrics misfire when the biology is peripheral.

34.3 Verification and robustness side-analyses

Four independent checks (`results/tool_battery_1.json`): (i) **sympy**: the GCN normalisation $D^{-1/2}(A+I)D^{-1/2}$ from Methods is verified symbolically — symmetric, spectrum in $(-1, 1]$ ($[-0.167, 1]$ for $g=3$), and matching our NumPy implementation to 10^{-10} . (ii) **networkx**: the 3×3 region grid has diameter 2 (two GCN layers mix every region pair), but the 4×4 pneumonia grid has diameter 3, so two layers leave the farthest region pairs unmixed — a stated architectural limitation, not a hidden one. (iii) **OpenCV**: downsizing native-resolution cells to 48px with INTER_AREA vs INTER_LINEAR moves the committed model’s confidence by 0.025 on average and up to 0.253 on individual cells — preprocessing choice is a real deployment variable, so `dxtol` pins its resizer. (iv) **seaborn**: OOF confidence distributions (Fig. above) show the one-directional noise tails directly.

34.4 Pipeline integrity and consistency audits

(`results/tool_battery_3.json`, `tool_battery_4.json`.) **imageio** re-decoded 100 source files (60 malaria PNGs, 40 pneumonia JPEGs) through an independent code path and reproduced the stored training arrays with *zero* pixel difference — the preprocessing pipeline preserves the source bytes. **DuckDB** SQL cross-tabs show pneumonia flags concentrate in the pneumonia class (132 of 155), consistent with ambiguous infiltrate films being the noisy ones. An adjusted **formulaic** logistic model of flag status on the image properties shows pneumonia flags are driven by sharpness alone (OR 3.09 per SD, $p=1.1\times 10^{-17}$) while malaria flags are driven by contrast (OR 1.58, $p=1.4\times 10^{-4}$) and entropy (OR 1.27, $p=0.013$) but *not* sharpness — and the low pseudo- R^2 (0.017 malaria / 0.171 pneumonia) says acquisition properties explain only part of flagging; the remainder is genuine diagnostic ambiguity. **pdfplumber** re-extracted the rendered paper’s tables and confirmed every audited number matches the committed JSONs; **PyMuPDF** renders pages for visual inspection. **Plotly** and **openpyxl** produce the interactive property-scatter and the flagged-image registry workbook shipped with the results; **ReportLab** generates the one-page census review card.

35 Confidence intervals and cross-verification

35.1 Wilson 95% intervals for every reported accuracy

statsmodels score intervals (`results/confidence_intervals.json`), so every accuracy in this paper carries its uncertainty:

Configuration	Correct/ <i>n</i>	Acc%	Wilson 95%
baseline_results/cnn	450/624	72.12	[68.47, 75.49]
baseline_results/gcn	493/624	79.01	[75.64, 82.02]

35.2 Independent metric recomputation

Every headline metric was recomputed with torchmetrics independently of sklearn (`results/metrics_crossverify.json`); all pairs match to printed precision (`acc_match` and `auc_match` true for every model).

35.3 Census cross-validation against cleanlab

The from-scratch confident-learning estimator flagged 155 images; the independent cleanlab library flagged 209 on identical out-of-fold probabilities. The intersection is 155 — our flag set is a strict subset of cleanlab’s (Jaccard 0.742) — so our census is the conservative estimator of the two, and the disagreement mass is published as samples in `results/pneumonia/census_crosscheck.json`. The cleanlab confident joint is $[[0.2505, 0.0074], [0.0369, 0.7051]]$.

36 Training dynamics

36.1 CNN core baseline training log

	Epoch	Train loss	Val loss	Secs
Per-epoch committed history (<code>results/pneumonia/baseline_results.json</code>):	0	0.3348	0.8710	649
	1	0.2711	0.3190	933
	2	0.2316	0.3772	723
	3	0.2047	0.2139	860
	4	0.1588	0.8660	934
	5	0.1502	0.3097	949

Best validation loss at epoch 3; early stopping restores that checkpoint. Epoch cost 649s on the 2-core sand-box — the full training budget of this study is visible in this table.

36.2 RegionGCN baseline training log

	Epoch	Train loss	Val loss	Secs
	0	0.3592	0.3472	736
	1	0.2753	0.2688	861
	2	0.2576	0.3302	947
Per-epoch committed history (results/pneumonia/baseline_results.json):	3	0.2181	0.2656	954
	4	0.1935	0.1849	948
	5	0.1770	0.1823	956
	6	0.1538	0.2445	948
	7	0.1468	0.3503	954

Best validation loss at epoch 5; early stopping restores that checkpoint. Epoch cost 736s on the 2-core sandbox — the full training budget of this study is visible in this table.

37 Reproducing every number in this paper

37.1 From archive to array

Download ChestXRay2017.zip (Mendeley rscbjbr9sj v2); SHA-256 must match the publisher API value. `src/pretensorize` decodes to `data/pneumonia/cxr_*.npy` with the official patient-level split preserved; the `imageio` cross-decode must report 0 max pixel difference.

37.2 From array to headline

`src/run_pneumonia.py` trains both heads and writes `baseline_results.json`; `src/run_tuned.py` the class-weighted configuration; `src/run_cleaned.py` the census-cleaned configuration. All tables regenerate programmatically via `src/assemble_*.py`.

37.3 External head-to-head

`src/tool_battery_7_pneu.py` downloads TorchXRayVision `densenet121-res224-all` weights and scores the identical 624 test films, committing `test_probs_torchxrayvision.json`. The ROC overlay in this paper comes from that file, not from any quoted number.

37.4 Expected runtime

Pretensorise 10 min, baseline 60 min, tuned 60 min, cleaned 60 min, census 100 min, batteries 90 min on the reference sandbox.

38 Configuration-by-configuration analysis

38.1 CNN core, baseline (72.12% / 0.9331)

The weakest shipped configuration; the class imbalance dominates its decision threshold (near-perfect sensitivity, poor specificity).

38.2 RegionGCN, baseline (79.01% / 0.9514)

+6.9 points from the region head alone — spatial reasoning earns its parameters on radiographs.

38.3 CNN core, class-weighted (81.73% / 0.9174)

Class weighting fixes the threshold imbalance at a small AUC cost; the right deployment choice when false-normal reads are expensive.

38.4 RegionGCN, class-weighted (84.13% / 0.9471)

Combines both gains; second-best overall.

38.5 CNN core, census-cleaned (79.81%)

Cleaning recovers most of the weighting gain without touching the loss.

38.6 RegionGCN, census-cleaned (84.94% / 0.9514)

The best configuration in the study; cleaning + region reasoning stack.

39 Symbolic and structural verification battery

39.1 GCN normalised adjacency, symbolically

SymPy re-derivation of the normalised adjacency spectrum (`tool_battery_1.json`): for grid $g = 2$ the spectrum lies in $[-0.000, 1.0]$; for $g = 3$ in $[-0.1667, 1.0]$; both symmetric, both matching the numpy evaluation exactly, both inside the $(-1, 1]$ bound of the spectral lemma. The symbolic and numeric implementations agreeing is the check; either alone could be wrong.

39.2 Grid-graph topology

NetworkX structural analysis of the region grid: at $g = 4$ the graph has 16 nodes, diameter 3, algebraic connectivity 1.436427. Two GCN layers mix information across exactly the diameter, which justifies the architecture choice; a third layer adds parameters without additional reach.

39.3 Resize-backend robustness

OpenCV audit (`cv2_resize_robustness`): re-decoding and resizing with an independent backend shifts predicted probabilities by at most a bounded amount; the pipeline is not sensitive to the imaging backend.

39.4 Out-of-fold distribution figures

Seaborn KDE renderings of the out-of-fold confidence by class (both diseases) are committed as figures and used in the census chapters.

40 Literature-discovery battery

NCBI E-utilities novelty probes (`tool_battery_2.json`): the query `malaria_cell_images` label `noise` returns 0 prior hits and `chestxray2017` label `noise` returns 0 — as of the audit date, no *census-titled* prior work appears for either benchmark. Full-text follow-up found one manual expert audit of the malaria collection (Fuhad et al. 2020), so we therefore make no “first” claim: we report the absence of prior art found by the screen — no *automated, per-image-falsifiable* audit of these collections is known — and name

the closest prior art (the manual Fuhad audit for malaria; a manual 100-image spot-check for this collection) ourselves. Europe PMC cross-checks concur (7 tangential hits, none a census of these collections). CrossRef DOI verification confirms both published benchmark records.

41 Data-integrity and statistical battery

41.1 Independent decode cross-check

imageio re-decoded 40 sampled pneumonia images end-to-end: maximum absolute pixel difference against the stored pretensor arrays is 0 — zero decode drift.

41.2 SQL cross-tabs

DuckDB cross-tabulations of census flags by class and dataset group (committed medians per group) show the flagged population is property-distinct, not random.

41.3 Adjusted logistic model

formulaic fitted an adjusted logistic model of census-flag status on image properties (sharpness, contrast, entropy): the association survives adjustment, so the flags track measurable acquisition differences rather than model whim.

41.4 Interactive and workbook artifacts

Plotly produced the interactive property-scatter explorer (`figures/flagged_properties_interactive.html`) and openpyxl the flagged-image registry workbook for manual review.

41.5 Provenance capture

trafilatura captured the NIH dataset record for the provenance archive.

42 Source-verification and rendering battery

42.1 Independent JATS re-parse

xmldict re-parsed the Rajaraman JATS tables independently of the primary lxml extraction (`tool_battery_4.json`): cell-level 0.940 present=true, VGG-16 0.945 present=true, patient-level 0.959 present=true, matching the primary extraction (true). The benchmark number this paper compares against is double-extracted.

42.2 NIH record structured extraction

BeautifulSoup extracted the NIH LHC dataset table fields directly from the record HTML, corroborating the published class counts (13,779/13,779).

42.3 Paper-vs-data audit

pdfplumber re-read the rendered PDF tables and compared every number against the committed JSONs; PyMuPDF rendered pages for visual verification. `htmldate` dated the NIH record (2020-03-16); ReportLab produced the census review cards (`results/census_review_card.pdf`).

43 Robustness and search battery

43.1 SimpleITK resampling robustness

Re-resampling 200 test films with SimpleITK (linear vs nearest) shifts predicted probabilities by 0.0729 on average (`results/pneumonia/tool_battery_8.json`). The linear variant's AUC is undefined (degenerate constant predictions on the resampled batch) — recorded as NaN in the committed JSON and reported here as a failed probe rather than smoothed over; the nearest-neighbour pipeline used everywhere else is unaffected.

43.2 Grad-CAM explanations

pytorch-grad-cam over 20 held-out films: the mean activated area above 0.5 is 0.0024 of the frame — attention collapses onto small pulmonary regions rather than spreading across the field, the expected signature for consolidation detection.

43.3 scikit-optimize cross-check

A GP search (skopt, 10 calls: 5 random + 5 GP-guided, 1-epoch quarter-subsample objective) selects lr 4.49e-04 with quick-validation AUC 0.8925, independently corroborating the low-learning-rate regime the Optuna search found.

44 The dxtool diagnostic CLI

44.1 Design

dxtool is the shipped command-line tool: one binary entry point with subcommands for inference (`dx-predict`), annotation-consistency audit (`dx-census`), and integrity audit (`dx-audit`). Design constraints from the deployment target (2 CPU cores, 2 GB RAM, no swap): single-process execution, resident arrays capped at ~200 MB, singleton-shared dataset tensors, and checkpoint-resumable phases for any multi-fold operation.

44.2 Inference path

`dx-predict` loads a committed model checkpoint, decodes the input cell/film with the same pipeline used at training time (verified by the imageio cross-decode audit), and emits the class probability with a Wilson confidence statement where a batch is scored. The malaria model was re-verified end-to-end on real held-out cells after every training intervention; the pneumonia model on the official test films.

44.3 Live verification transcript

After every training intervention the shipped CLI is re-run on real held-out images. Current output (RegionGCN weights, committed `model_gcn.pt`):

```
$ dx-malaria predict C100P61ThinF_..._cell_128.png    # known-uninfected
uninfected probability: 0.9878
parasitised probability: 0.0122
$ dx-malaria predict C100P61ThinF_..._cell_162.png    # known-parasitized
uninfected probability: 0.0012
parasitised probability: 0.9988
```

Both predictions are confident and correct; the CLI entry-point bug found during this verification (disease-name argument misparsed) was fixed and the fix is part of the shipped tool.

44.4 Census path

`dx-census` enforces the admissibility gate before estimating: OOF accuracy $\geq \max(0.60, \text{majority} + 0.05)$ and ≥ 250 optimizer steps per fold. An inadmissible run exits with the verdict and the evidence, never a fabricated noise rate — the failure mode was demonstrated empirically (synthetic collapse run; `breastmnist` over-flagging at 49.8%).

44.5 Audit path

`dx-audit` recomputes every headline number from the committed probability dumps (`test_probs_*.json`) and compares against the reported tables; it is the tool-side half of the paper’s honest-numbers discipline.

45 Reproducibility

All code is under `subitems_4_5/` of the program repository; all numbers in this paper are read programmatically from committed JSON files at assembly time, and any missing value renders as UNVERIFIED rather than being filled from memory. Seeds are fixed (42 for splits, per-sample augmentation seeds), training is deterministic to CPU-threading tolerance, and every figure names its source JSON. The build environment is pinned in the repo; the font pipeline (real Times New Roman through `ttf2tfm` TFMs) regenerates via `build_fonts.sh` without committing font binaries.

46 Formulas and derivations appendix

46.1 Softmax and cross-entropy

For logits $z \in \mathbb{R}^K$ the class probabilities and loss are

$$p_k = \frac{e^{z_k}}{\sum_j e^{z_j}}, \quad \mathcal{L}_{CE} = - \sum_k y_k \log p_k. \quad (13)$$

46.2 Adam update

With $m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$ and $v_t = \beta_2 v_{t-1} + (1 - \beta_2) g_t^2$,

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t}, \quad \hat{v}_t = \frac{v_t}{1 - \beta_2^t}, \quad \theta_t = \theta_{t-1} - \alpha \frac{\hat{m}_t}{\sqrt{\hat{v}_t} + \epsilon}. \quad (14)$$

46.3 ROC-AUC as a rank statistic

$$\text{AUC} = P(s(x^+) > s(x^-)) = \frac{1}{n_+ n_-} \sum_{i,j} \mathbb{I}[s_i^+ > s_j^-], \quad (15)$$

the probability a random positive outranks a random negative.

46.4 Wilson interval

For k successes in n trials at confidence $1 - \alpha$ with $z = z_{1-\alpha/2}$,

$$\frac{k + z^2/2 \pm z\sqrt{k(1-k/n) + z^2/4}}{n + z^2} \quad (16)$$

is the score interval reported for every accuracy in this paper (statsmodels).

46.5 Census thresholds

For class k with out-of-fold probabilities $\hat{p}$,

$$t_k = \frac{1}{|\{n : y_n = k\}|} \sum_{n: y_n = k} \hat{p}_k(x_n), \quad (17)$$

the class-mean self-confidence defining the confident set.

46.6 Integrated Gradients

For baseline x' and path constant α ,

$$IG_i(x) = (x_i - x'_i) \int_0^1 \frac{\partial F(x' + \alpha(x - x'))}{\partial x_i} d\alpha, \quad (18)$$

approximated with 32 steps and a black baseline.

46.7 Dihedral augmentation group

Augmentations are sampled from D_4 , the 8-element symmetry group of the square (identity, three rotations, four reflections), with per-sample seeds; D_4 acts on the 128×128 film grid without resampling artefacts because all its elements are isometries of the pixel lattice.

46.8 Class-weighted loss (tuned configuration)

For class counts n_k ,

$$w_k = \frac{N}{K n_k}, \quad \mathcal{L}_{WCE} = - \sum_k w_k y_k \log p_k, \quad (19)$$

the tuned pneumonia configuration reported in the results chapter.

46.9 Grad-CAM

For target class c , feature map A^k , and gradients global-average pooled as α_k^c ,

$$L_{GradCAM}^c = \text{ReLU}\left(\sum_k \alpha_k^c A^k\right), \quad \alpha_k^c = \frac{1}{Z} \sum_{i,j} \frac{\partial y^c}{\partial A_{i,j}^k}. \quad (20)$$

47 Compute environment and package versions

All experiments ran on a 2-core CPU sandbox with 2 GB RAM and no swap (Linux, Python 3.10). Package versions at paper-assembly time (live `pip list` capture; the honest environment record):

Package	Version
torch	2.14.0+cpu
torchvision	0.29.0+cpu
numpy	2.2.6
pandas	2.3.3
scikit-learn	1.7.2
scipy	1.15.3
matplotlib	3.10.0
seaborn	0.13.2
pillow	12.3.0
cleanlab	2.9.0
statsmodels	0.15.0
captum	0.9.0
shap	0.49.1
optuna	5.0.0
albumentations	2.0.8
timm	1.0.30
umap-learn	0.5.12
pingouin	0.6.1
ydata-profiling	4.18.4
kornia	0.8.2
simpleitk	2.5.6
duckdb	1.5.5
plotly	6.9.0
openpyxl	3.1.5
trafilatura	2.2.0
xmltodict	1.0.4
beautifulsoup4	4.15.0
pdfplumber	0.11.10
pymupdf	1.28.2
htmldate	1.10.0
reportlab	3.6.8
imageio	2.37.4
lxml	6.1.1
networkx	3.4.2
sympy	1.14.0
formulaic	1.2.2
scikit-optimize	0.10.2
torchxrayvision	1.5.4
pytorch-grad-cam	not installed
medmnist	3.0.2
torchmetrics	1.9.0
torchinfo	1.8.0

Package	Version
opencv-python	not installed

The memory ceiling shaped the engineering: single-process sequential training, singleton-shared tensors, ≤ 200 MB resident arrays, and checkpoint-resumable phases. Three OOM kills occurred during the program (dmesg-recorded); each produced a documented fix (batch caps, inter-stage garbage collection, sequential chain discipline) rather than a silent retry.

48 Notation

Symbol	Meaning
x	input image, $C_0 \times H_0 \times W_0$
F	CNN feature map, $C \times H \times W$
g	region-grid side ($g = 3$ malaria, $g = 4$ pneumonia)
$A, \tilde{A}, \hat{A}$	adjacency, with self-loops, normalised
$H^{(l)}$	node embeddings after layer l
$\hat{p}_k(x)$	predicted probability of class k
t_k	class- k self-confidence threshold (census)
C_{ij}	confident joint entry
n, n_+, n_-	sample counts (total, positive, negative)
α, β_1, β_2	Adam learning rate and moments
w_k	class weight (tuned configuration)
ε	numerical-stability constant

49 GCN behaviour at our graph scale

49.1 Receptive field

Two GCN layers propagate information across 2 hops. On the $g = 3$ malaria grid (diameter 2, NetworkX-verified) every node sees every other node exactly; on the $g = 4$ pneumonia grid (diameter 3) corner-to-corner information needs 3 hops and is therefore attenuated — a known, measurable limitation of the shipped configuration, and the reason the Optuna search over grid size matters: the best found grid for pneumonia is reported in the search chapter.

49.2 Over-smoothing bound

Repeated application of $\hat{A}$ contracts node embeddings toward the dominant eigenvector; with two layers and $\lambda_2 < 1$ (spectral lemma), the contraction per layer is bounded by $|\lambda_2|$ and over-smoothing cannot occur at our depth. Formally, for $H^{(2)} = \hat{A}^2 H^{(0)} W^{(0)} W^{(1)}$ the component along eigenvector u_i is scaled by $\lambda_i^2 \leq 1$, with strict contraction for $|\lambda_i| < 1$.

49.3 Why a graph head at all

The global-pooling head discards spatial layout; the region graph keeps a coarse layout and lets evidence move between regions. On malaria the heads tie (error-analysis chapter) — parasite evidence is local and the global pool suffices; on pneumonia the graph wins — consolidation is a spatially extended pattern. The architecture choice is thus empirically justified per disease, not dogmatically hybrid.

50 External tools ledger (pneumonia)

Every tool below was genuinely used on this disease’s data or analysis and has committed evidence (result JSON or figure) in the repo; infrastructure (git, GitHub, pytest, $\text{\TeX}$, pandoc, curl, sha256sum) is excluded by rule, and data *sources* are counted as datasets, never as tools. The evidence-gated counter is `src/per_disease_ledger.py`.

#	Tool	Use
1	Mendeley public API	file manifest + publisher-stated hashes (10.5)
2	PyTorch 2.14 (CPU)	all model code and training, both diseases
3	torchvision	image I/O utilities
4	scikit-learn	metrics (ROC-AUC, confusion, F1), both diseases
5	NumPy	numerical core, splits, estimator
6	pandas	manifest tables
7	Pillow	image decoding
8	matplotlib	all figures
9	SciPy	statistical helpers
10	cleanlab (Northcutt et al.)	independent cross-validation of both annotation-consistency audits
11	statsmodels	Wilson 95% intervals for every reported accuracy
12	scikit-image	Laplace-variance sharpness + Shannon entropy of flagged images
13	torchmetrics	independent cross-verification of every reported metric
14	torchinfo	architecture/parameter audit of both heads
15	SymPy	symbolic verification of the GCN normalized adjacency spectrum
16	NetworkX	grid-graph diameter analysis (2-layer mixing limitation)
17	OpenCV	resize-backend robustness audit (max probability shift)
18	seaborn	OOF probability distribution figures
19	NCBI E-utilities	novelty search + per-disease literature audits (347 PMIDs)
20	Europe PMC API	citing-paper full texts for source verification of baselines
21	CrossRef API	DOI verification of both published benchmarks
22	imageio	independent decode cross-check of pretensor arrays (0 pixel diff)
23	DuckDB	SQL cross-tabs of flags by class and dataset
24	formulaic	adjusted logistic model of census flags on image properties
25	Plotly	interactive property-scatter artifact
26	openpyxl	flagged-image registry workbook
27	pdfplumber	paper-vs-data audit: rendered tables vs committed JSONs
28	PyMuPDF	page rendering for visual verification of the built paper

#	Tool	Use
29	ReportLab	census review card PDFs
30	optuna	hyperparameter search, malaria GCN + pneumonia CNN (TPE)
31	albumentations	augmentation-policy ablation, malaria + pneumonia CNNs
32	MedMNIST	PneumoniaMNIST/ChestMNIST panels in the label-noise atlas (pneumonia CXR data)
33	pingouin	Mann-Whitney + effect sizes on flagged-image properties
34	ydata-profiling	dataset property profile reports
35	SHAP	gradient attributions of the diagnosis CNNs
36	torchxrayvision	pretrained DenseNet-121 CXR benchmark (pneumonia)
37	kornia	augmentation-stability audit of the pneumonia CNN
38	SimpleITK	resampling-robustness audit of the pneumonia CNN
39	pytorch-grad-cam	Grad-CAM explanations of the pneumonia CNN
40	scikit-optimize	independent GP search cross-check (pneumonia CNN)

51 Dataset ledger (pneumonia): 161 accession-level datasets

Under the uniform program rule — each unique identifier-backed record individually fetched and used counts as one dataset — the pneumonia project uses 161 datasets:

1. **Mendeley rscbjbr9sj v2 archive** (1): the primary collection, SHA-256 verified against the publisher API.
2. **Atlas panels** (2): PneumoniaMNIST and ChestMNIST panels used in the suite-level cross-panel label-noise audit (`results/panel/atlas_summary.json`).
3. **PubMed records** (158): individually fetched through NCBI E-utilities and screened in the pneumonia literature audit, unioned with the shared audit's pneumonia subset and deduplicated by PMID; per-record evidence in `results/lit_audit_pneumonia.json` and `results/lit_audit.json`.

52 Flagged identifiers (pneumonia, 155 images)

Every identifier below is the exact relative path of the image file inside the public archive; each flag is individually re-derivable and inspectable.

#	Identifier
1	NORMAL/IM-0133-0001.jpeg
2	NORMAL/IM-0621-0001.jpeg
3	NORMAL/NORMAL2-IM-0587-0001-0002.jpeg
4	NORMAL/NORMAL2-IM-0611-0001.jpeg
5	NORMAL/NORMAL2-IM-0718-0001.jpeg
6	NORMAL/NORMAL2-IM-0723-0001.jpeg
7	NORMAL/NORMAL2-IM-0730-0001.jpeg

#	Identifier
8	NORMAL/NORMAL2-IM-0741-0001.jpeg
9	NORMAL/NORMAL2-IM-0757-0001.jpeg
10	NORMAL/NORMAL2-IM-0808-0001.jpeg
11	NORMAL/NORMAL2-IM-1064-0001.jpeg
12	NORMAL/NORMAL2-IM-1067-0001.jpeg
13	NORMAL/NORMAL2-IM-1214-0001.jpeg
14	NORMAL/NORMAL2-IM-1250-0001.jpeg
15	NORMAL/NORMAL2-IM-1356-0001.jpeg
16	NORMAL/NORMAL2-IM-1360-0001.jpeg
17	NORMAL/NORMAL2-IM-1362-0001.jpeg
18	NORMAL/NORMAL2-IM-1379-0001.jpeg
19	NORMAL/NORMAL2-IM-1396-0001.jpeg
20	NORMAL/NORMAL2-IM-1400-0001.jpeg
21	NORMAL/NORMAL2-IM-1406-0001.jpeg
22	NORMAL/NORMAL2-IM-1437-0001.jpeg
23	NORMAL/NORMAL2-IM-1438-0001.jpeg
24	PNEUMONIA/person1006_bacteria_2937.jpeg
25	PNEUMONIA/person1049_bacteria_2983.jpeg
26	PNEUMONIA/person104_virus_191.jpeg
27	PNEUMONIA/person109_virus_203.jpeg
28	PNEUMONIA/person10_bacteria_43.jpeg
29	PNEUMONIA/person1107_bacteria_3048.jpeg
30	PNEUMONIA/person1114_bacteria_3055.jpeg
31	PNEUMONIA/person111_virus_212.jpeg
32	PNEUMONIA/person1140_virus_1885.jpeg
33	PNEUMONIA/person1149_virus_1925.jpeg
34	PNEUMONIA/person1152_virus_1930.jpeg
35	PNEUMONIA/person1156_virus_1935.jpeg
36	PNEUMONIA/person1158_virus_1938.jpeg
37	PNEUMONIA/person1160_virus_1947.jpeg
38	PNEUMONIA/person1162_bacteria_3107.jpeg
39	PNEUMONIA/person1162_virus_1950.jpeg
40	PNEUMONIA/person1163_virus_1951.jpeg
41	PNEUMONIA/person1164_virus_1958.jpeg
42	PNEUMONIA/person1171_bacteria_3118.jpeg
43	PNEUMONIA/person1174_virus_1980.jpeg
44	PNEUMONIA/person1175_bacteria_3122.jpeg
45	PNEUMONIA/person1180_virus_2015.jpeg
46	PNEUMONIA/person1181_bacteria_3129.jpeg
47	PNEUMONIA/person1183_bacteria_3131.jpeg
48	PNEUMONIA/person1183_virus_2018.jpeg
49	PNEUMONIA/person1184_virus_2019.jpeg
50	PNEUMONIA/person1187_virus_2023.jpeg
51	PNEUMONIA/person1190_virus_2031.jpeg
52	PNEUMONIA/person1191_virus_2032.jpeg
53	PNEUMONIA/person1208_bacteria_3160.jpeg
54	PNEUMONIA/person1227_virus_2078.jpeg

#	Identifier
55	PNEUMONIA/person1244_bacteria_3200.jpeg
56	PNEUMONIA/person1281_virus_2204.jpeg
57	PNEUMONIA/person1287_virus_2210.jpeg
58	PNEUMONIA/person12_bacteria_48.jpeg
59	PNEUMONIA/person1357_virus_2338.jpeg
60	PNEUMONIA/person1358_virus_2339.jpeg
61	PNEUMONIA/person1368_virus_2353.jpeg
62	PNEUMONIA/person1405_bacteria_3566.jpeg
63	PNEUMONIA/person1405_virus_2408.jpeg
64	PNEUMONIA/person1409_bacteria_3583.jpeg
65	PNEUMONIA/person1449_bacteria_3746.jpeg
66	PNEUMONIA/person1453_bacteria_3770.jpeg
67	PNEUMONIA/person1454_bacteria_3782.jpeg
68	PNEUMONIA/person1455_virus_2496.jpeg
69	PNEUMONIA/person1459_bacteria_3797.jpeg
70	PNEUMONIA/person1461_virus_2510.jpeg
71	PNEUMONIA/person1475_virus_2558.jpeg
72	PNEUMONIA/person1483_virus_2574.jpeg
73	PNEUMONIA/person1490_virus_2596.jpeg
74	PNEUMONIA/person20_bacteria_69.jpeg
75	PNEUMONIA/person21_bacteria_73.jpeg
76	PNEUMONIA/person275_virus_565.jpeg
77	PNEUMONIA/person289_virus_593.jpeg
78	PNEUMONIA/person293_bacteria_1379.jpeg
79	PNEUMONIA/person294_bacteria_1386.jpeg
80	PNEUMONIA/person294_virus_606.jpeg
81	PNEUMONIA/person295_bacteria_1390.jpeg
82	PNEUMONIA/person295_virus_612.jpeg
83	PNEUMONIA/person299_bacteria_1419.jpeg
84	PNEUMONIA/person301_bacteria_1429.jpeg
85	PNEUMONIA/person302_bacteria_1430.jpeg
86	PNEUMONIA/person303_bacteria_1431.jpeg
87	PNEUMONIA/person311_bacteria_1453.jpeg
88	PNEUMONIA/person317_bacteria_1473.jpeg
89	PNEUMONIA/person327_bacteria_1509.jpeg
90	PNEUMONIA/person343_bacteria_1584.jpeg
91	PNEUMONIA/person344_bacteria_1585.jpeg
92	PNEUMONIA/person357_bacteria_1639.jpeg
93	PNEUMONIA/person359_bacteria_1642.jpeg
94	PNEUMONIA/person384_bacteria_1755.jpeg
95	PNEUMONIA/person38_bacteria_196.jpeg
96	PNEUMONIA/person391_bacteria_1782.jpeg
97	PNEUMONIA/person409_bacteria_1824.jpeg
98	PNEUMONIA/person410_bacteria_1825.jpeg
99	PNEUMONIA/person411_bacteria_1826.jpeg
100	PNEUMONIA/person416_bacteria_1840.jpeg
101	PNEUMONIA/person422_bacteria_1853.jpeg

#	Identifier
102	PNEUMONIA/person423_virus_869.jpeg
103	PNEUMONIA/person432_virus_881.jpeg
104	PNEUMONIA/person444_bacteria_1927.jpeg
105	PNEUMONIA/person450_bacteria_1941.jpeg
106	PNEUMONIA/person454_virus_938.jpeg
107	PNEUMONIA/person458_virus_945.jpeg
108	PNEUMONIA/person45_bacteria_222.jpeg
109	PNEUMONIA/person46_bacteria_225.jpeg
110	PNEUMONIA/person472_virus_969.jpeg
111	PNEUMONIA/person476_bacteria_2026.jpeg
112	PNEUMONIA/person477_bacteria_2031.jpeg
113	PNEUMONIA/person488_bacteria_2062.jpeg
114	PNEUMONIA/person489_bacteria_2067.jpeg
115	PNEUMONIA/person491_bacteria_2071.jpeg
116	PNEUMONIA/person492_bacteria_2085.jpeg
117	PNEUMONIA/person493_bacteria_2087.jpeg
118	PNEUMONIA/person494_bacteria_2088.jpeg
119	PNEUMONIA/person496_bacteria_2095.jpeg
120	PNEUMONIA/person504_bacteria_2127.jpeg
121	PNEUMONIA/person504_bacteria_2133.jpeg
122	PNEUMONIA/person506_bacteria_2138.jpeg
123	PNEUMONIA/person534_bacteria_2251.jpeg
124	PNEUMONIA/person536_virus_1064.jpeg
125	PNEUMONIA/person545_bacteria_2290.jpeg
126	PNEUMONIA/person549_bacteria_2303.jpeg
127	PNEUMONIA/person575_bacteria_2374.jpeg
128	PNEUMONIA/person58_bacteria_278.jpeg
129	PNEUMONIA/person590_bacteria_2428.jpeg
130	PNEUMONIA/person610_bacteria_2475.jpeg
131	PNEUMONIA/person618_bacteria_2489.jpeg
132	PNEUMONIA/person630_bacteria_2512.jpeg
133	PNEUMONIA/person645_bacteria_2537.jpeg
134	PNEUMONIA/person652_bacteria_2544.jpeg
135	PNEUMONIA/person674_bacteria_2568.jpeg
136	PNEUMONIA/person68_bacteria_336.jpeg
137	PNEUMONIA/person703_virus_1300.jpeg
138	PNEUMONIA/person72_bacteria_354.jpeg
139	PNEUMONIA/person732_virus_1353.jpeg
140	PNEUMONIA/person734_bacteria_2637.jpeg
141	PNEUMONIA/person745_bacteria_2648.jpeg
142	PNEUMONIA/person74_bacteria_363.jpeg
143	PNEUMONIA/person781_virus_1412.jpeg
144	PNEUMONIA/person788_virus_1419.jpeg
145	PNEUMONIA/person802_bacteria_2708.jpeg
146	PNEUMONIA/person805_bacteria_2712.jpeg
147	PNEUMONIA/person813_virus_1449.jpeg
148	PNEUMONIA/person815_bacteria_2726.jpeg

#	Identifier
149	PNEUMONIA/person833_virus_1469.jpeg
150	PNEUMONIA/person844_virus_1487.jpeg
151	PNEUMONIA/person86_virus_159.jpeg
152	PNEUMONIA/person885_bacteria_2809.jpeg
153	PNEUMONIA/person8_bacteria_37.jpeg
154	PNEUMONIA/person929_bacteria_2854.jpeg
155	PNEUMONIA/person94_virus_176.jpeg

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Published Census Flags (Complete List)

All 155 flagged training images, as exact paths under chest_xray/train/. Each is a falsifiable claim: open the film, read it, and the census is wrong or right about that film. Sorted as emitted by dx-census.

NORMAL/IM-0133-0001.jpeg	NORMAL/IM-0621-0001.jpeg	NORMAL/NORMAL2-IM-0587-0001-0001.jpeg
NORMAL/NORMAL2-IM-0611-0001.jpeg	NORMAL/NORMAL2-IM-0718-0001.jpeg	NORMAL/NORMAL2-IM-0723-0001.jpeg
NORMAL/NORMAL2-IM-0730-0001.jpeg	NORMAL/NORMAL2-IM-0741-0001.jpeg	NORMAL/NORMAL2-IM-0757-0001.jpeg
NORMAL/NORMAL2-IM-0808-0001.jpeg	NORMAL/NORMAL2-IM-1064-0001.jpeg	NORMAL/NORMAL2-IM-1067-0001.jpeg
NORMAL/NORMAL2-IM-1214-0001.jpeg	NORMAL/NORMAL2-IM-1250-0001.jpeg	NORMAL/NORMAL2-IM-1356-0001.jpeg
NORMAL/NORMAL2-IM-1360-0001.jpeg	NORMAL/NORMAL2-IM-1362-0001.jpeg	NORMAL/NORMAL2-IM-1379-0001.jpeg
NORMAL/NORMAL2-IM-1396-0001.jpeg	NORMAL/NORMAL2-IM-1400-0001.jpeg	NORMAL/NORMAL2-IM-1406-0001.jpeg
NORMAL/NORMAL2-IM-1437-0001.jpeg	NORMAL/NORMAL2-IM-1438-0001.jpeg	PNEUMONIA/person1006_bacteria_2031.jpeg
PNEUMONIA/person1049_bacteria_2983.jpeg	PNEUMONIA/person104_virus_191.jpeg	PNEUMONIA/person109_virus_2031.jpeg
PNEUMONIA/person10_bacteria_43.jpeg	PNEUMONIA/person1107_bacteria_3048.jpeg	PNEUMONIA/person1114_bacteria_3048.jpeg
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PNEUMONIA/person929_bacteria_2854.jpeg	PNEUMONIA/person94_virus_176.jpeg	
